NUST institute of Civil Engineering (NICE), National University of Sciences and Technology (NUST),

Sector H-12, Islamabad.

Design of Concrete Structures

Complex Engineering Problem

Bachelors Hostel

Group 02

Group Members:

Muhammad Umer Basit (259084)
Haziq Ahmad (243247)
Muhammad Talha Nadeem (249311)
Syed Haider Ali Sherazi (246730)
Muhammad Saad (259022)
Ahmer Naseer (244724)

Table of Content

1. **Chapter 01 Introduction**
   - 1.1. Problem Statement
   - 1.2. Objective
   - 1.3. Assumptions
2. **Chapter 02 Design Criteria**
   - 2.1. Dead Loads
   - 2.2. Live Loads
   - 2.3. Earthquake Forces
   - 2.4. Mass Source
   - 2.5. Load Combinations
3. **Chapter 03 Modelling Procedure**
   - 3.1. Architectural Layout
   - 3.2 Modelling of the Structure
4. **Chapter 04 Analysis Procedure**
5. **Chapter 05 Structural Design**
   - 5.1. Design Methodology
   - 5.2. Material Property
   - 5.3. Structural Proportioning
   - 5.4. Structural System
6. **Chapter 6 Detailing Rules**
7. **Summary**
8. **Appendices**

1. Introduction:

Structural design is the primary aspect of civil engineering. The very basis of construction of any building, residential house or dams, bridges, culverts, canals etc. is designing. The aim is to provide a safe, serviceable, durable, economical and aesthetically pleasing structure. For the structure to be safe, it must be able to resist the worst loading conditions. Under normal working conditions, the deformation and cracking must not be excessive for the structure to remain serviceable, durable and aesthetically pleasing during the expected design life. Furthermore, the structure should be economical with regard to both construction and maintenance cost.

The foremost task in structural engineering is the design of simple basic components and members of a building like slabs, beams, columns and footings. For their design and architectural design is created on Auto CAD. In order to design them it is important to first obtain the plan of the particular building that is positioning of particular rooms such that they serve their respective purpose and also suiting to the requirements and comfort of the inhabitants. Thereby depending on the suitability, plan layout of beams and the columns are fixed. Thereafter, the vertical loads are calculated namely the dead load and the live load from ASCE 7-16.

1.1. Problem Statement:

A bachelor's hostel, with 2x persons in each room having attach bath facility. The hostel is a 4 storey with a capacity of 500 students. There must be dining, recreational and prayer room of suitable dimensions. The hostel is proposed to be constructed on an area of 150'x200'. There is also an underground water tank of capacity of 10,000 gallons. After the completion of preliminary design in accordance with code based design procedures, the performance of the building is checked for serviceable behavior for frequent earthquakes (10% Probability of exceedance in 50 years).

1.2. Objective:

During designing, a sensible designer ought to bear in mind that structure should be a balancing of economy, aesthetic and stability. One can always design a massive structure, which has more than adequate stability strength and serviceability, but the ensuring cost of the structure may be exorbitant and the end product far from aesthetics. In the design of structures, the aim is to design the structure in such a way that it fulfils its intended purpose during its intended life time and be adequately safe in terms of strength, stability and structural rigidity and have adequate serviceability in terms of stiffness, durability etc.

Safety requires the possibility of collapse of the structure (partial or total) is acceptably low not under normal expected loads (service loads), but also under less frequent loads (such as due to earthquakes). Other two important considerations that a sensible designer ought to bear in mind are that the structure should be economical with regard to both construction and maintenance cost and aesthetically pleasing during the expected design life.

1.3. Assumptions:

The site was assumed to be in Quetta, Balochistan. The Spectral acceleration values used were of this site and designed accordingly. Soil stresses was taken from ASCE 7-16 simply because of no available data. The values of super imposed dead loads were assumed by considering mortar screeding and tiles as surface finishing which are acting as permanent loads. Site class was assumed to be D and the soil profile assumed was stiff. Occupancy Importance Factor was assumed 1. The retaining walls were assumed to be shear walls.

2. Design Criteria:

Building standards and loadings were taken from ACI 318-14 and ASCE 7-16. The following design cases were enveloped in design:

Load Calculation:

The various loads considered for analysis were:-

2.1. Dead loads

The dimensions of the cross section are to be assumed initially which enable to estimate the dead load from the known weights of the structure. The values of the unit weights of the structure. The values of the unit weight of the materials are specified in ASCE 7-16

- Unit weight of brick = 130lb/ft^3.
- Unit weight of concrete = 150lb/ft^3.
- Wall load Thickness of wall = 9inch.
- Main wall load = 750lb/ft.
- Parapet wall load = 340lb/ft.
- Staircase Flight Slab = 47.96lb/ft^2.
- Floor Finish Load = 16lb/ft^2.
- Roof Load = 16.08lb/ft^2.

2.2. Live Loads:

The live load values are also provided in ASCE 7-16 but the lateral hydraulic pressure was determined by taking product of unit weight of water 62.4lb/ft^3 and the depth of tank 5ft.

- Occupancy Load = 36lb/ft^2.
- Water Tank Load = 312lb/ft^2.
- Soil Load = 35lb/ft^2.

2.3. Earthquake forces:

Earthquakes generate waves which move from the origin of its location with velocities depending on the intensity and magnitude of the earthquake. The impact of earthquake on the structures depends on the stiffness of the structure, stiffness of the soil media, height and location of the structure, etc. the earthquake forces are prescribed in ASCE 7-16. Since the building is located in Quetta, Balochistan. It is included in the zone 2-B. And the seismic base shear calculation and its distribution was done as ASCE 7-16. The Spectral acceleration taken for the site for DBE of 10% probability of exceedance in 50 years were 0.7g for 0.2 seconds time period Ss, and 0.25g for 1 seconds time period S1. The site class was D. Fa and Fv values were determined by ETABS using D site class. Long period transition period was taken as 8 seconds. Response modification factors R, System overstrength Ω_o, Deflection amplification factor Cd were taken as prescribed in ASCE 7-16 table according to the provisions provided for Ordinary Reinforced Concrete Moment Resisting Frame. Occupancy Importance Factor was taken as 1.

2.4. Mass Source:

Seismic weight for the calculation of base shear was also taken from ASCE 7-16.

$$V = C_s W$$

Where C_s is the seismic coefficient and W is the effective seismic weight.

This seismic effective weight included all the dead loads and 25% of the total imposed live loads as per ASCE 7-16 provisions.

2.5. Load Combinations:

Default cases already present in ETABS were used to design the structural elements as per ACI 318-14 and ASCE 7-16. These cases included were for frame elements, slabs and the walls that were modelled for the underground water tank. Load patterns defined in the ETABS all are incorporated in the respective load cases and designed for.

3. Modelling:

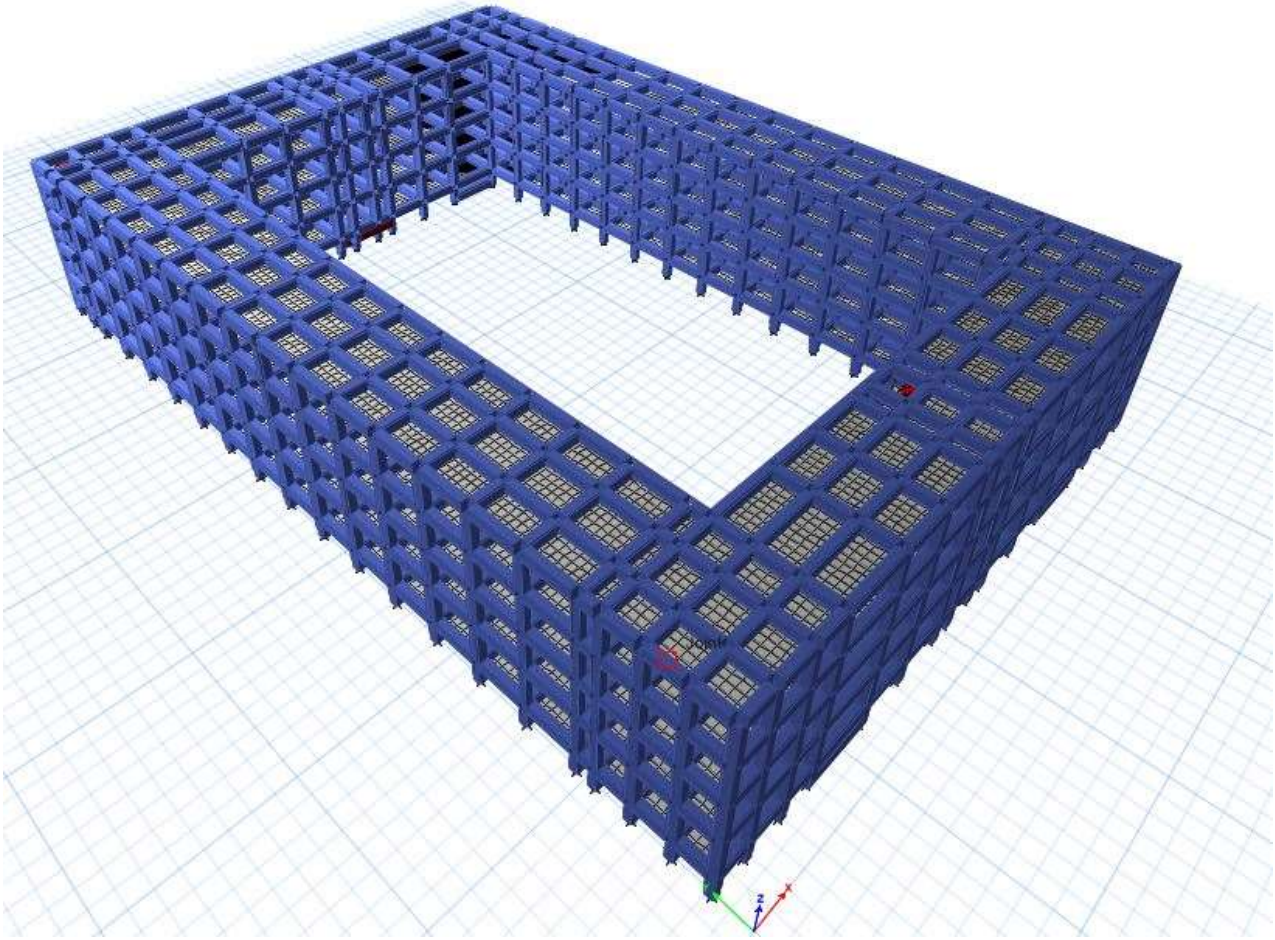
3.1. Architectural Layout:

There was a need for accommodation of 500 people and a constraint on number of storeys so rooms were so dimensioned so as to accommodate 2 people each with attached baths. Underground water tank also was needed to be designed. The inspiration was taken from NUST hostels and so design of the hostel was an imitation of these hostels with recreation spaces in the central area of the hostel. One staircase was provided at entrance and two at the back for ease of accessibility. Mess hall alongside kitchen was provided at the back in Ground Floor. Gym area is right above the mess hall at first floor. Praying areas are provided at the front corners of the ground floor.

i. Type of building – Residential building.
ii. Type of structure –multi storey rigid jointed framed.
iii. No. of storey – 4 (G+3).
iv. Plinth to First floor height – 12ft.
v. Floor to floor height – 11ft.
vi. External walls – 9inch.
vii. Internal walls – 9inch.
viii. Height of plinth – 7ft

3.2. Modelling of the structure:

ETABS was used for modelling of the structure. Grid system for the infrastructure of the frame structure which consists of elements like beams and columns was created. After the conceived grid lines the columns were placed at the junctions of grid lines. Subsequent elements like beams were placed afterwards. Staircase was modelled thereafter. The flight slabs were provided and two waist slabs were provided of width half of the flight slabs for the staircase. Roof Slabs were created at Plinth level and the upper stories with openings at staircases and central part reserved for recreational area. For underwater tank, the depth was 5ft and the surface area of 14x21 ft^2 for holding a maximum of 10,000 gal. Walls were modelled against the hydraulic and the soil pressure acting laterally on them. Meshing is done to break the large shell elements like slabs and shear walls into small elements to develop a precise, correct and uniform load distribution pattern from slabs to beams. There are two ways to do the assigning: Manual Method and Auto-Mesh. The latter is generally adopted which was also used in this modelling.



4. Analysis Procedure:

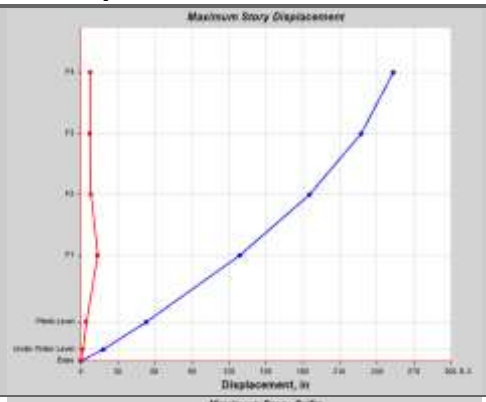
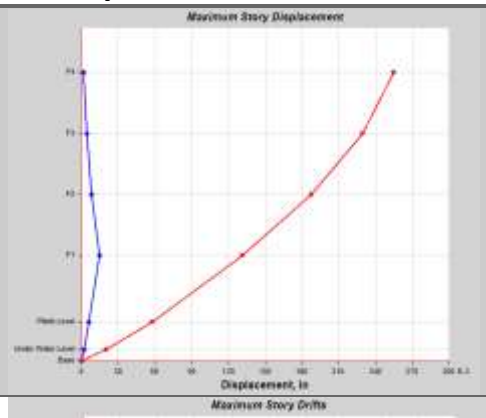
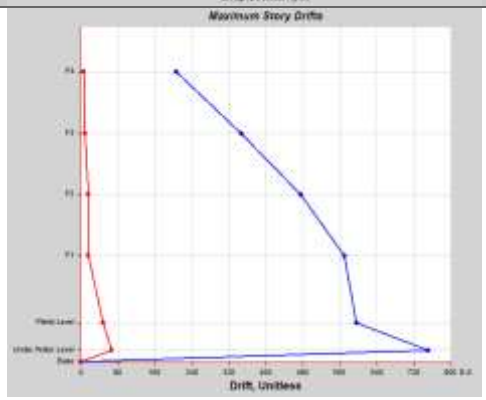
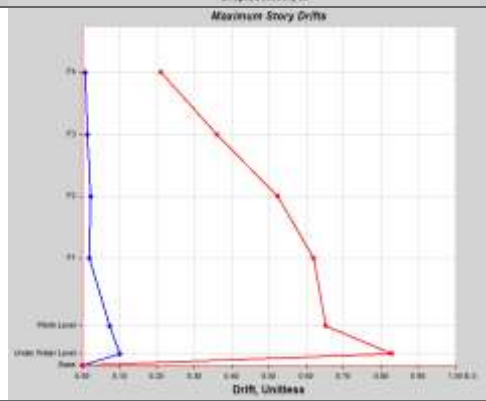
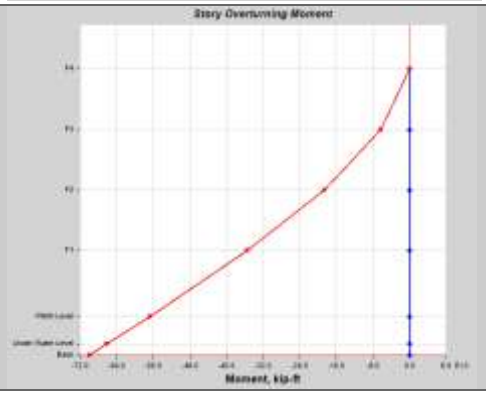
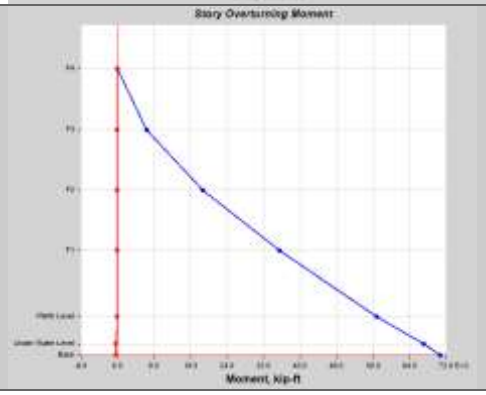
The structure was analyzed as ordinary moment resisting space frames in the versatile software ETABS. From the analysis, maximum design loads, moments and shear on each member was obtained. From these values, we design the structure. Modal analysis was performed to check the stability and adequacy of the model. There was no torsion in the first two modes. The table 01 shows the modal periods and frequencies from the modal analysis of the structure.

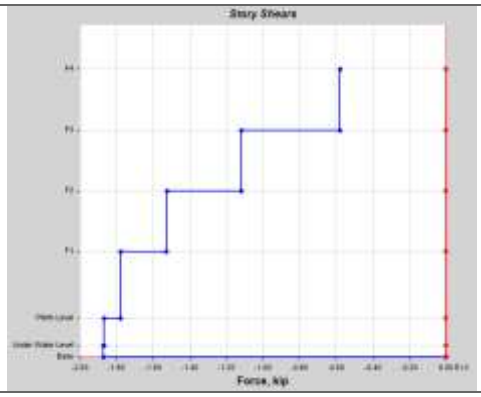
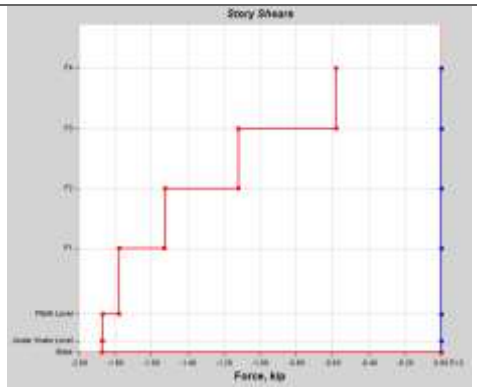
TABLE 01: Modal Periods And frequencies

Case	Mode	Period	Frequency	CircFreq	Eigenvalue
		sec	cyc/sec	rad/sec	rad²/sec²
Modal	1	0.473	2.116	13.2937	176.7214
Modal	2	0.465	2.151	13.5141	182.6299
Modal	3	0.441	2.268	14.2523	203.1286
Modal	4	0.323	3.097	19.4578	378.6077
Modal	5	0.259	3.862	24.2667	588.8725

Modal	6	0.171	5.857	36.8006	1354.2833
Modal	7	0.146	6.873	43.1834	1864.802
Modal	8	0.145	6.876	43.2033	1866.5234
Modal	9	0.142	7.065	44.3926	1970.7051
Modal	10	0.138	7.233	45.448	2065.5202
Modal	11	0.132	7.554	47.4608	2252.532
Modal	12	0.126	7.954	49.9771	2497.7147

Seismic Analysis results are also shown in the following table which show the story drifts and displacements which were well within the permissible limits of said parameters.

Seismic Results	Earthquake in X-Direction	Earthquake in Y-Direction
Maximum Storey Displacements		
Maximum Storey Drifts		
Storey Overturning Moments		

Storey Shears		

5. **Structural Design:**
6. **Design Methodology:**



6.1. Material Property:

The materials used for design were concrete and grade 60 steel. The concrete locally available was of 3000 psi strength. These material properties greatly influenced our choices for the sections chosen and consequently the design. The stiffness modifiers were also added: 0.7 for columns and 0.35 for beams. Torsional constant was 0.01 for both of them. The modifiers was used to model the actual real behavior of concrete under sustained loading. Cracking occurs in the concrete when it reaches tensile capacity, which is about 7 to 10% of the compressive strength. The moment of inertia is reduced hence moment capacity also reduced so we have to reduce our stiffness and thus causes more deflection.

6.2. Structure Proportioning:

Trial cross-sections were assumed initially. Beams of 18x24 in^2 and columns of 18x18 in^2. The design by using these section showed that they were inadequate and the shear due to the combination of shear and torsion exceeded the sections' capacities even when reinforced to maximum permissible steel reinforcements as prescribed by ACI 318-14. The sections were modified to 24x24 in^2 for beams and the columns to same. Two internal columns underneath the underwater tank were failing in the same manner so they had to be increased to 30x30 in^2. A set of columns were very closely placed and the beams resting on them were failing so these redundant columns were removed from the model.

6.3. Structural System:

Because of the overall height of the structure, we have opted for Ordinary RC Moment Resisting Frame; it consists of two columns (vertical elements) and one beam (horizontal element) connecting the tops of the columns. Specifically, a frame makes the connections between the beams and columns rigid. For RC, that means there is reinforcing steel that bends from the column to the beam, and the concrete for the beam is intended to bond with the concrete in the column. Both continuity of reinforcement and concrete are required to make the frame resist moments.

7. Detailing Rules:

The bar size chosen for the beams and columns was #6. For the slabs it was #5. The reason was the provisions of ACI 318-14 and local availability. The rest of specifications like curtailment length were the default options in ETABS. For extensive detailing visualization for the model, CSI Detail, a plugin for ETABS was used. The drawing are provided.

Bill of Quantities:

Using CSI Detail, the Bill of Quantities of different frame elements is presented consisting of steel rebars and concrete. All the BOQs are attached.

8. Summary:

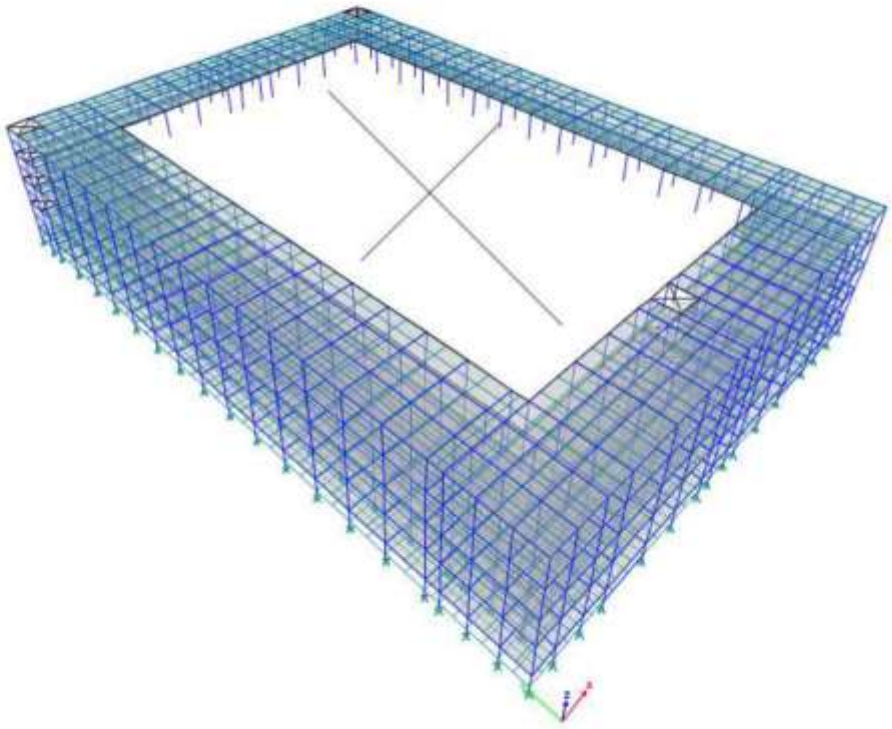
After analysing the G+3 storey building structure, concluded that structure is safe in loading like dead load, live load and seismic load. Member dimensions (Beam, Column, Slab and Walls) are assigned by calculating the load type and its quantity applied on it. CSI Detail plan gives detailed information of the structural members length, height, depth, size & numbers etc. CSI ETABS has the capability to

calculate the reinforcement needed for any concrete section. The program contains a number of parameters which are designed as per ACI 318-14 and ASCE 7-16. Beams are designed for flexure, shear and torsion and their combined actions and it gives the detail number, position, spacing of reinforcement. The analysis and design was done according to standard specifications to the possible extend. The various difficulties encountered in the design process and the various constraints faced by the structural engineer in designing up to the architectural drawing were also understood.

Appendix A The design reports from ETABS is provided with this file as follows.

Appendix B The drawings from CSI Detail and ETABS is also attached.

ETABS®



Bachelors Hotel

Model Group 02, Revision 0
1/9/2022

1 Properties

This chapter provides property information for materials, frame sections, shell sections, and links. 1.1 Materials

Table 1.1 - Material Properties - General

Material	Type	SymType	Grade	Color	Notes
3000Psi	Concrete	Isotropic	f'c 3000 psi	Gray8Dark	
A416Gr270	Tendon	Uniaxial	Grade 270	Green	
A615Gr60	Rebar	Uniaxial	Grade 60	Blue	
A992Fy50	Steel	Isotropic	Grade 50	Yellow	

1.2 Frame Sections

Table 1.2 - Frame Section Property Definitions - Summary (Part 1 of 3)

Name	Material	Shape	Color	Area in2	J in4in4	I33	I22 in4	As2 in2	As3 in2
A-CompBm		Auto Select							
A-GravBm		Auto Select							
A-GravCol		Auto Select							
A-LatBm		Auto Select							
A-LatCol		Auto Select							
Chota Kaduu Column	3000Psi	Concrete Rectangular	Cyan	900	114075	67500	67500	750	750
ConcBm	3000Psi	Concrete Rectangular	Green	576	46725.12	27648	27648	480	480
ConcCol	3000Psi	Concrete Rectangular	Blue	576	46725.12	27648	27648	480	480
SteelBm	A992Fy50	Steel I/Wide Flange	Gray8Dark	10.25	0.59	561.85	18.02	4.49	5.58
SteelCol	A992Fy50	Steel I/Wide Flange	Yellow	44	12.71	2774.67	972.17	9.04	32.04
W10X100	A992Fy50	Steel I/Wide Flange	Cyan	29.3	10.9	623	207	7.55	19.23
W10X112	A992Fy50	Steel I/Wide Flange	Green	32.9	15.1	716	236	8.61	21.67
W10X12	A992Fy50	Steel I/Wide Flange	Gray8Dark	3.54	0.05	53.8	2.18	1.88	1.39
W10X15	A992Fy50	Steel I/Wide Flange	Yellow	4.41	0.1	68.9	2.89	2.3	1.8
W10X17	A992Fy50	Steel I/Wide Flange	Magenta	4.99	0.16	81.9	3.56	2.42	2.21
W10X19	A992Fy50	Steel I/Wide Flange	Red	5.62	0.23	96.3	4.29	2.55	2.65
W10X33	A992Fy50	Steel I/Wide Flange	Magenta	9.71	0.58	171	36.6	2.82	5.77
W10X39	A992Fy50	Steel I/Wide Flange	Red	11.5	0.98	209	45	3.12	7.06
W10X45	A992Fy50	Steel I/Wide Flange	Cyan	13.3	1.51	248	53.4	3.54	8.29
W10X49	A992Fy50	Steel I/Wide Flange	Green	14.4	1.39	272	93.4	3.4	9.33
W10X54	A992Fy50	Steel I/Wide Flange	Blue	15.8	1.82	303	103	3.74	10.25
W10X60	A992Fy50	Steel I/Wide Flange	Gray8Dark	17.7	2.48	341	116	4.28	11.45
W10X68	A992Fy50	Steel I/Wide Flange	Yellow	19.9	3.56	394	134	4.89	12.96
W10X77	A992Fy50	Steel I/Wide Flange	Magenta	22.7	5.11	455	154	5.62	14.79
W10X88	A992Fy50	Steel I/Wide Flange	Red	26	7.53	534	179	6.53	17
W12X106	A992Fy50	Steel I/Wide Flange	Red	31.2	9.13	933	301	7.87	20.13
W12X120	A992Fy50	Steel I/Wide Flange	Cyan	35.2	12.9	1070	345	9.3	22.76
W12X136	A992Fy50	Steel I/Wide Flange	Cyan	39.9	18.5	1240	398	10.59	25.83
W12X14	A992Fy50	Steel I/Wide Flange	Cyan	4.16	0.07	88.6	2.36	2.38	1.49
W12X152	A992Fy50	Steel I/Wide Flange	Green	44.7	25.8	1430	454	11.92	29.17
W12X16	A992Fy50	Steel I/Wide Flange	Green	4.71	0.1	103	2.82	2.64	1.76
W12X170	A992Fy50	Steel I/Wide Flange	Blue	50	35.6	1650	517	13.44	32.76

Table 1.2 - Frame Section Property Definitions - Summary (Part 1 of 3, continued)

Name	Material	Shape	Color	Area in2	J in4	I33 in4	I22 in4	As2 in2	As3 in2
W12X19	A992Fy50	Steel I/Wide Flange	Blue	5.57	0.18	130	3.76	2.87	2.34
W12X190	A992Fy50	Steel I/Wide Flange	Gray8Dark	56	48.8	1890	589	15.26	36.83
W12X210	A992Fy50	Steel I/Wide Flange	Yellow	61.8	64.7	2140	664	17.35	40.53
W12X22	A992Fy50	Steel I/Wide Flange	Gray8Dark	6.48	0.29	156	4.66	3.2	2.85
W12X26	A992Fy50	Steel I/Wide Flange	Cyan	7.65	0.3	204	17.3	2.81	4.11
W12X30	A992Fy50	Steel I/Wide Flange	Green	8.79	0.46	238	20.3	3.2	4.78
W12X40	A992Fy50	Steel I/Wide Flange	Blue	11.7	0.91	307	44.1	3.51	6.88
W12X45	A992Fy50	Steel I/Wide Flange	Blue	13.1	1.26	348	50	4.05	7.71
W12X50	A992Fy50	Steel I/Wide Flange	Gray8Dark	14.6	1.71	391	56.3	4.51	8.62
W12X53	A992Fy50	Steel I/Wide Flange	Yellow	15.6	1.58	425	95.8	4.17	9.58
W12X58	A992Fy50	Steel I/Wide Flange	Magenta	17	2.1	475	107	4.39	10.67
W12X65	A992Fy50	Steel I/Wide Flange	Red	19.1	2.18	533	174	4.72	12.1
W12X72	A992Fy50	Steel I/Wide Flange	Cyan	21.1	2.93	597	195	5.29	13.4
W12X79	A992Fy50	Steel I/Wide Flange	Green	23.2	3.84	662	216	5.83	14.82
W12X87	A992Fy50	Steel I/Wide Flange	Blue	25.6	5.1	740	241	6.44	16.34
W12X96	A992Fy50	Steel I/Wide Flange	Magenta	28.2	6.85	833	270	6.99	18.3
W14X109	A992Fy50	Steel I/Wide Flange	Gray8Dark	32	7.12	1240	447	7.51	20.93
W14X120	A992Fy50	Steel I/Wide Flange	Yellow	35.3	9.37	1380	495	8.56	23.03
W14X132	A992Fy50	Steel I/Wide Flange	Magenta	38.8	12.3	1530	548	9.48	25.24
W14X145	A992Fy50	Steel I/Wide Flange	Red	42.7	15.2	1710	677	10.06	28.16
W14X159	A992Fy50	Steel I/Wide Flange	Cyan	46.7	19.7	1900	748	11.18	30.94
W14X176	A992Fy50	Steel I/Wide Flange	Green	51.8	26.5	2140	838	12.62	34.28
W14X193	A992Fy50	Steel I/Wide Flange	Blue	56.8	34.8	2400	931	13.8	37.68
W14X211	A992Fy50	Steel I/Wide Flange	Gray8Dark	62	44.6	2660	1030	15.39	41.08
W14X22	A992Fy50	Steel I/Wide Flange	Yellow	6.49	0.21	199	7	3.15	2.79
W14X233	A992Fy50	Steel I/Wide Flange	Yellow	68.5	59.5	3010	1150	17.12	45.58
W14X257	A992Fy50	Steel I/Wide Flange	Magenta	75.6	79.1	3400	1290	19.35	50.4
W14X26	A992Fy50	Steel I/Wide Flange	Magenta	7.69	0.36	245	8.91	3.54	3.52
W14X283	A992Fy50	Steel I/Wide Flange	Red	83.3	104	3840	1440	21.54	55.55
W14X30	A992Fy50	Steel I/Wide Flange	Blue	8.85	0.38	291	19.6	3.73	4.32
W14X311	A992Fy50	Steel I/Wide Flange	Cyan	91.4	136	4330	1610	24.11	61.02
W14X34	A992Fy50	Steel I/Wide Flange	Gray8Dark	10	0.57	340	23.3	3.99	5.12
W14X342	A992Fy50	Steel I/Wide Flange	Green	101	178	4900	1810	26.95	67.51
W14X370	A992Fy50	Steel I/Wide Flange	Blue	109	222	5440	1990	29.71	73.15
W14X38	A992Fy50	Steel I/Wide Flange	Yellow	11.2	0.8	385	26.7	4.37	5.81
W14X398	A992Fy50	Steel I/Wide Flange	Gray8Dark	117	273	6000	2170	32.39	78.85
W14X426	A992Fy50	Steel I/Wide Flange	Yellow	125	331	6600	2360	35.16	84.61
W14X43	A992Fy50	Steel I/Wide Flange	Gray8Dark	12.6	1.05	428	45.2	4.18	7.07
W14X455	A992Fy50	Steel I/Wide Flange	Magenta	134	395	7190	2560	38.38	89.88
W14X48	A992Fy50	Steel I/Wide Flange	Yellow	14.1	1.45	484	51.4	4.69	7.96
W14X500	A992Fy50	Steel I/Wide Flange	Red	147	514	8210	2880	42.92	99.17
W14X53	A992Fy50	Steel I/Wide Flange	Magenta	15.6	1.94	541	57.7	5.14	8.87
W14X61	A992Fy50	Steel I/Wide Flange	Red	17.9	2.19	640	107	5.21	10.75
W14X68	A992Fy50	Steel I/Wide Flange	Cyan	20	3.01	722	121	5.81	12
W14X74	A992Fy50	Steel I/Wide Flange	Yellow	21.8	3.87	795	134	6.39	13.21

Table 1.2 - Frame Section Property Definitions - Summary (Part 1 of 3, continued)

Name	Material	Shape	Color	Area in2	J in4	I33 in4	I22 in4	As2 in2	As3 in2
W14X82	A992Fy50	Steel I/Wide Flange	Magenta	24	5.07	881	148	7.29	14.39
W14X90	A992Fy50	Steel I/Wide Flange	Green	26.5	4.06	999	362	6.16	17.16
W14X99	A992Fy50	Steel I/Wide Flange	Blue	29.1	5.37	1110	402	6.89	18.98
W16X26	A992Fy50	Steel I/Wide Flange	Magenta	7.68	0.26	301	9.59	3.93	3.16
W16X31	A992Fy50	Steel I/Wide Flange	Red	9.13	0.46	375	12.4	4.37	4.06
W16X36	A992Fy50	Steel I/Wide Flange	Cyan	10.6	0.55	448	24.5	4.69	5.01
W18X35	A992Fy50	Steel I/Wide Flange	Cyan	10.3	0.51	510	15.3	5.31	4.25
W18X40	A992Fy50	Steel I/Wide Flange	Green	11.8	0.81	612	19.1	5.64	5.27
W18X46	A992Fy50	Steel I/Wide Flange	Blue	13.5	1.22	712	22.5	6.52	6.11
W18X50	A992Fy50	Steel I/Wide Flange	Blue	14.7	1.24	800	40.1	6.39	7.13
W18X55	A992Fy50	Steel I/Wide Flange	Gray8Dark	16.2	1.66	890	44.9	7.06	7.91
W18X60	A992Fy50	Steel I/Wide Flange	Yellow	17.6	2.17	984	50.1	7.55	8.76
W21X44	A992Fy50	Steel I/Wide Flange	Green	13	0.77	843	20.7	7.25	4.88
W21X50	A992Fy50	Steel I/Wide Flange	Gray8Dark	14.7	1.14	984	24.9	7.9	5.82
W21X55	A992Fy50	Steel I/Wide Flange	Magenta	16.2	1.24	1140	48.4	7.8	7.15
W21X57	A992Fy50	Steel I/Wide Flange	Red	16.7	1.77	1170	30.6	8.55	7.11
W21X62	A992Fy50	Steel I/Wide Flange	Red	18.3	1.83	1330	57.5	8.4	8.45
W21X68	A992Fy50	Steel I/Wide Flange	Cyan	20	2.45	1480	64.7	9.07	9.44
W24X55	A992Fy50	Steel I/Wide Flange	Yellow	16.2	1.18	1350	29.1	9.32	5.9
W24X62	A992Fy50	Steel I/Wide Flange	Green	18.2	1.71	1550	34.5	10.19	6.92
W24X68	A992Fy50	Steel I/Wide Flange	Green	20.1	1.87	1830	70.4	9.84	8.75
W24X76	A992Fy50	Steel I/Wide Flange	Blue	22.4	2.68	2100	82.5	10.52	10.19
W24X84	A992Fy50	Steel I/Wide Flange	Gray8Dark	24.7	3.7	2370	94.4	11.33	11.58
W24X94	A992Fy50	Steel I/Wide Flange	Yellow	27.7	5.26	2700	109	12.51	13.23
W27X102	A992Fy50	Steel I/Wide Flange	Red	30	5.28	3620	139	13.96	13.83
W27X114	A992Fy50	Steel I/Wide Flange	Cyan	33.6	7.33	4080	159	15.56	15.66
W27X129	A992Fy50	Steel I/Wide Flange	Green	37.8	11.1	4760	184	16.84	18.33
W27X84	A992Fy50	Steel I/Wide Flange	Blue	24.7	2.81	2850	106	12.28	10.67
W27X94	A992Fy50	Steel I/Wide Flange	Magenta	27.6	4.03	3270	124	13.18	12.42
W30X108	A992Fy50	Steel I/Wide Flange	Blue	31.7	4.99	4470	146	16.24	13.3
W30X116	A992Fy50	Steel I/Wide Flange	Gray8Dark	34.2	6.43	4930	164	16.95	14.88
W30X124	A992Fy50	Steel I/Wide Flange	Yellow	36.5	7.99	5360	181	17.67	16.28
W30X132	A992Fy50	Steel I/Wide Flange	Magenta	38.8	9.72	5770	196	18.63	17.5
W30X90	A992Fy50	Steel I/Wide Flange	Magenta	26.3	2.84	3610	115	13.87	10.57
W30X99	A992Fy50	Steel I/Wide Flange	Red	29	3.77	3990	128	15.44	11.73
W33X118	A992Fy50	Steel I/Wide Flange	Cyan	34.7	5.3	5900	187	18.1	14.18
W33X130	A992Fy50	Steel I/Wide Flange	Red	38.3	7.37	6710	218	19.2	16.39
W33X141	A992Fy50	Steel I/Wide Flange	Cyan	41.5	9.7	7450	246	20.15	18.4
W33X152	A992Fy50	Steel I/Wide Flange	Green	44.9	12.4	8160	273	21.27	20.49
W36X135	A992Fy50	Steel I/Wide Flange	Green	39.9	7	7800	225	21.36	15.8
W36X150	A992Fy50	Steel I/Wide Flange	Blue	44.3	10.1	9040	270	22.44	18.8
W36X160	A992Fy50	Steel I/Wide Flange	Gray8Dark	47	12.4	9760	295	23.4	20.4
W36X170	A992Fy50	Steel I/Wide Flange	Yellow	50	15.1	10500	320	24.62	22
W36X182	A992Fy50	Steel I/Wide Flange	Yellow	53.6	18.5	11300	347	26.32	23.8
W8X31	A992Fy50	Steel I/Wide Flange	Red	9.13	0.54	110	37.1	2.28	5.8

Table 1.2 - Frame Section Property Definitions - Summary (Part 1 of 3, continued)

Name	Material	Shape	Color	Area in2	J in4	I33 in4	I22 in4	As2 in2	As3 in2
W8X35	A992Fy50	Steel I/Wide Flange	Cyan	10.3	0.77	127	42.6	2.52	6.62
W8X40	A992Fy50	Steel I/Wide Flange	Green	11.7	1.12	146	49.1	2.97	7.53
W8X48	A992Fy50	Steel I/Wide Flange	Blue	14.1	1.96	184	60.9	3.4	9.26
W8X58	A992Fy50	Steel I/Wide Flange	Gray8Dark	17.1	3.33	228	75.1	4.46	11.1
W8X67	A992Fy50	Steel I/Wide Flange	Yellow	19.7	5.05	272	88.6	5.13	12.9

Table 1.2 - Frame Section Property Definitions - Summary (Part 2 of 3)

S33Pos in3	S33Neg in3	S22Pos in3	S22Neg in3	Z33 in3	Z22 in3	R33 in	R22 in	Cw in6	Fillet Radius in	CG Offset 3 in	CG Offset 2 in	PNA Offset 3 in
4500	4500	4500	4500	6750	6750	8.6603	8.6603			0	0	0
2304	2304	2304	2304	3456	3456	6.9282	6.9282			0	0	0
2304	2304	2304	2304	3456	3456	6.9282	6.9282			0	0	0
62.43	62.43	6.01	6.01	70.56	9.27	7.4037	1.326	1378.13		0	0	0
308.3	308.3	108.02	108.02	338	163	7.9411	4.7005	70227		0	0	0
112.25	112.25	40.19	40.19	130	61	4.6112	2.658	5079.02	0.5	0	0	0
125.61	125.61	45.38	45.38	147	69.2	4.6651	2.6783	6035.74	0.5	0	0	0
10.9	10.9	1.1	1.1	12.6	1.74	3.8984	0.7847	50.7	0.3	0	0	0
13.79	13.79	1.45	1.45	16	2.3	3.9527	0.8095	68.02	0.3	0	0	0
16.22	16.22	1.78	1.78	18.7	2.8	4.0513	0.8446	84.63	0.3	0	0	0
18.88	18.88	2.13	2.13	21.6	3.35	4.1395	0.8737	102.79	0.3	0	0	0
35.15	35.15	9.2	9.2	38.8	14	4.1965	1.9415	789.8	0.5	0	0	0
42.14	42.14	11.26	11.26	46.8	17.2	4.2631	1.9781	993.2	0.5	0	0	0
49.11	49.11	13.32	13.32	54.9	20.3	4.3182	2.0038	1197.62	0.5	0	0	0
54.4	54.4	18.68	18.68	60.4	28.3	4.3461	2.5468	2079.32	0.5	0	0	0
60	60	20.6	20.6	66.6	31.3	4.3792	2.5532	2305.36	0.505	0	0	0
66.86	66.86	22.97	22.97	74.6	35	4.3893	2.56	2645.67	0.5	0	0	0
75.77	75.77	26.53	26.53	85.3	40.1	4.4496	2.5949	3065.46	0.5	0	0	0
85.85	85.85	30.2	30.2	97.6	45.9	4.4771	2.6046	3641.95	0.5	0	0	0
98.89	98.89	34.76	34.76	113	53.1	4.5319	2.6239	4337.84	0.5	0	0	0
144.65	144.65	49.34	49.34	164	75.1	5.4684	3.106	10624.95	0.6	0	0	0
163.36	163.36	56.1	56.1	186	85.4	5.5134	3.1307	12372.73	0.59	0	0	0
185.07	185.07	64.19	64.19	214	98	5.5747	3.1583	14659.41	0.6	0	0	0
14.89	14.89	1.19	1.19	17.4	1.9	4.615	0.7532	79.96	0.3	0	0	0
208.76	208.76	72.64	72.64	243	111	5.6561	3.1869	17236.82	0.6	0	0	0
17.17	17.17	1.41	1.41	20.1	2.26	4.6764	0.7738	96.59	0.3	0	0	0
235.71	235.71	82.06	82.06	275	126	5.7446	3.2156	20121.75	0.6	0	0	0
21.31	21.31	1.88	1.88	24.7	2.98	4.8311	0.8216	132.05	0.3	0	0	0
262.5	262.5	92.76	92.76	311	143	5.8095	3.2431	23802.17	0.59	0	0	0
291.16	291.16	103.75	103.75	348	159	5.8845	3.2779	27201.46	0.6	0	0	0

Table 1.2 - Frame Section Property Definitions - Summary (Part 2 of 3, continued)

S33Pos in3	S33Neg in3	S22Pos in3	S22Neg in3	Z33 in3	Z22 in3	R33 in	R22 in	Cw in6	Fillet Radius in	CG Offset 3 in	CG Offset 2 in	PNA Offset 3 in
25.37	25.37	2.31	2.31	29.3	3.66	4.9065	0.848	163.44	0.3	0	0	0
33.44	33.44	5.33	5.33	37.2	8.17	5.164	1.5038	604.7	0.3	0	0	0
38.7	38.7	6.23	6.23	43.1	9.56	5.2035	1.5197	714.75	0.3	0	0	0
51.6	51.6	11.01	11.01	57	16.8	5.1224	1.9415	1429.42	0.505	0	0	0
57.52	57.52	12.42	12.42	64.2	19	5.1541	1.9537	1660.07	0.505	0	0	0
64.1	64.1	13.94	13.94	71.9	21.3	5.175	1.9637	1879.83	0.5	0	0	0
70.25	70.25	19.16	19.16	77.9	29.1	5.2195	2.4781	3182.28	0.605	0	0	0
77.87	77.87	21.4	21.4	86.4	32.5	5.2859	2.5088	3563.56	0.6	0	0	0
88.1	88.1	29	29	96.8	44.1	5.2826	3.0183	5755.8	0.595	0	0	0
97.07	97.07	32.5	32.5	108	49.2	5.3192	3.04	6524.79	0.6	0	0	0
106.77	106.77	35.7	35.7	119	54.3	5.3418	3.0513	7382.47	0.595	0	0	0
118.4	118.4	39.83	39.83	132	60.4	5.3765	3.0682	8170.69	0.6	0	0	0
131.18	131.18	44.26	44.26	147	67.5	5.435	3.0943	9481.45	0.6	0	0	0
173.43	173.43	61.23	61.23	192	92.7	6.225	3.7375	20143.94	0.6	0	0	0
190.34	190.34	67.35	67.35	212	102	6.2525	3.7447	22876.42	0.6	0	0	0
208.16	208.16	74.56	74.56	234	113	6.2796	3.7582	25475.05	0.6	0	0	0
231.08	231.08	87.35	87.35	260	133	6.3283	3.9818	31789.61	0.6	0	0	0
253.33	253.33	95.9	95.9	287	146	6.3785	4.0021	35900.2	0.6	0	0	0
281.58	281.58	106.75	106.75	320	163	6.4275	4.0221	40753.37	0.6	0	0	0
309.68	309.68	118.6	118.6	355	180	6.5003	4.0486	45900.86	0.6	0	0	0
338.85	338.85	130.38	130.38	390	198	6.5501	4.0759	51260.57	0.6	0	0	0
29.05	29.05	2.8	2.8	33.2	4.39	5.5374	1.0385	311.66	0.4	0	0	0
376.25	376.25	144.65	144.65	436	221	6.6288	4.0974	58744.2	0.6	0	0	0
414.63	414.63	161.25	161.25	487	246	6.7062	4.1308	67911.81	0.6	0	0	0
35.25	35.25	3.54	3.54	40.2	5.54	5.6444	1.0764	404.69	0.4	0	0	0
459.88	459.88	178.88	178.88	542	274	6.7896	4.1578	77041.62	0.6	0	0	0
42.17	42.17	5.82	5.82	47.3	8.99	5.7342	1.4882	879.99	0.4	0	0	0
506.43	506.43	198.77	198.77	603	304	6.8829	4.197	88167.81	0.6	0	0	0
48.57	48.57	6.9	6.9	54.6	10.6	5.831	1.5264	1069.72	0.4	0	0	0
560	560	220.73	220.73	672	338	6.9653	4.2333	102549.89	0.6	0	0	0
607.82	607.82	241.21	241.21	736	370	7.0646	4.2728	115635.76	0.6	0	0	0
54.61	54.61	7.89	7.89	61.5	12.1	5.863	1.544	1228.8	0.4	0	0	0
655.74	655.74	261.45	261.45	801	402	7.1611	4.3066	129662.64	0.59	0	0	0
705.88	705.88	282.63	282.63	869	434	7.2664	4.3451	144675.59	0.59	0	0	0
62.48	62.48	11.3	11.3	69.6	17.3	5.8282	1.894	1961.13	0.59	0	0	0
756.84	756.84	304.76	304.76	936	468	7.3251	4.3709	158119.67	0.6	0	0	0
70.14	70.14	12.8	12.8	78.4	19.6	5.8589	1.9093	2238.36	0.595	0	0	0
837.76	837.76	338.82	338.82	1050	522	7.4733	4.4263	185718.56	0.6	0	0	0
77.84	77.84	14.32	14.32	87.1	22	5.8889	1.9232	2524.14	0.59	0	0	0
92.09	92.09	21.4	21.4	102	32.8	5.9795	2.4449	4721.8	0.595	0	0	0
103.14	103.14	24.2	24.2	115	36.9	6.0083	2.4597	5290.75	0.59	0	0	0
111.97	111.97	26.53	26.53	126	40.5	6.0389	2.4793	6064.62	0.595	0	0	0
123.22	123.22	29.31	29.31	139	44.8	6.0587	2.4833	6635	0.595	0	0	0
142.71	142.71	49.93	49.93	157	75.6	6.1399	3.696	15929.46	0.6	0	0	0

Table 1.2 - Frame Section Property Definitions - Summary (Part 2 of 3, continued)

S33Pos in3	S33Neg in3	S22Pos in3	S22Neg in3	Z33 in3	Z22 in3	R33 in	R22 in	Cw in6	Fillet Radius in	CG Offset 3 in	CG Offset 2 in	PNA Offset 3 in
156.34	156.34	55.07	55.07	173	83.6	6.1761	3.7168	18215.75	0.6	0	0	0
38.34	38.34	3.49	3.49	44.2	5.48	6.2604	1.1175	563.89	0.402	0	0	0
47.17	47.17	4.48	4.48	54	7.03	6.4089	1.1654	741.03	0.402	0	0	0
56.35	56.35	7.01	7.01	64	10.8	6.5011	1.5203	1464.43	0.402	0	0	0
57.63	57.63	5.1	5.1	66.5	8.06	7.0367	1.2188	1141.48	0.402	0	0	0
68.38	68.38	6.35	6.35	78.4	10	7.2017	1.2723	1440.75	0.402	0	0	0
78.67	78.67	7.43	7.43	90.7	11.7	7.2623	1.291	1717.08	0.405	0	0	0
88.89	88.89	10.69	10.69	101	16.6	7.3771	1.6516	3043.98	0.402	0	0	0
98.34	98.34	11.93	11.93	112	18.5	7.412	1.6648	3420.58	0.4	0	0	0
108.13	108.13	13.25	13.25	123	20.6	7.4772	1.6872	3834.1	0.405	0	0	0
81.45	81.45	6.37	6.37	95.4	10.2	8.0527	1.2619	2111.5	0.5	0	0	0
94.62	94.62	7.63	7.63	110	12.2	8.1816	1.3015	2549.03	0.505	0	0	0
109.62	109.62	11.78	11.78	126	18.4	8.3887	1.7285	4967.35	0.498	0	0	0
110.9	110.9	9.33	9.33	129	14.8	8.3702	1.3536	3197.42	0.5	0	0	0
126.67	126.67	13.96	13.96	144	21.7	8.5251	1.7726	5957.54	0.505	0	0	0
140.28	140.28	15.65	15.65	160	24.4	8.6023	1.7986	6728.13	0.505	0	0	0
114.41	114.41	8.3	8.3	134	13.3	9.1287	1.3403	3866.07	0.505	0	0	0
130.8	130.8	9.8	9.8	153	15.7	9.2285	1.3768	4580.98	0.5	0	0	0
154.43	154.43	15.7	15.7	177	24.5	9.5417	1.8715	9399.61	0.505	0	0	0
175.73	175.73	18.35	18.35	200	28.6	9.6825	1.9191	11099.44	0.5	0	0	0
196.68	196.68	20.93	20.93	224	32.6	9.7955	1.955	12815.29	0.5	0	0	0
222.22	222.22	24.04	24.04	254	37.5	9.8728	1.9837	14927.18	0.505	0	0	0
267.16	267.16	27.8	27.8	305	43.4	10.9848	2.1525	23866.4	0.6	0	0	0
298.9	298.9	31.49	31.49	343	49.3	11.0195	2.1753	27762.34	0.6	0	0	0
344.93	344.93	36.8	36.8	395	57.6	11.2217	2.2063	32186.46	0.6	0	0	0
213.48	213.48	21.2	21.2	244	33.2	10.7417	2.0716	18109.96	0.6	0	0	0
243.12	243.12	24.8	24.8	278	38.8	10.8848	2.1196	21235.11	0.595	0	0	0
300	300	27.81	27.81	346	43.9	11.8747	2.1461	30914.59	0.65	0	0	0
328.67	328.67	31.24	31.24	378	49.2	12.0063	2.1898	34837.96	0.65	0	0	0
354.97	354.97	34.48	34.48	408	54	12.1181	2.2269	38431.3	0.65	0	0	0
380.86	380.86	37.33	37.33	437	58.4	12.1947	2.2476	41408.73	0.65	0	0	0
244.75	244.75	22.12	22.12	283	34.7	11.7159	2.0911	23862.38	0.65	0	0	0
268.69	268.69	24.38	24.38	312	38.6	11.7297	2.1009	27234.88	0.65	0	0	0
358.66	358.66	32.52	32.52	415	51.3	13.0395	2.3214	48500.48	0.7	0	0	0
405.44	405.44	37.91	37.91	467	59.5	13.2362	2.3858	56334.33	0.705	0	0	0
447.45	447.45	42.78	42.78	514	66.9	13.3984	2.4347	63625.84	0.7	0	0	0
487.16	487.16	47.07	47.07	559	73.9	13.481	2.4658	72548.81	0.7	0	0	0
438.24	438.23	37.5	37.5	509	59.7	13.9817	2.3747	68923.55	0.75	0	0	0
503.62	503.62	45	45	581	70.9	14.2851	2.4688	82718.6	0.75	0	0	0
542.22	542.22	49.17	49.17	624	77.3	14.4104	2.5053	89861.21	0.75	0	0	0
580.11	580.11	53.33	53.33	668	83.8	14.4914	2.5298	97575.19	0.75	0	0	0
622.59	622.59	57.36	57.36	718	90.7	14.5197	2.5444	107432.55	0.75	0	0	0
27.5	27.5	9.28	9.28	30.4	14.1	3.4711	2.0158	531.09	0.394	0	0	0
31.28	31.28	10.62	10.62	34.7	16.1	3.5114	2.0337	618.58	0.394	0	0	0

Table 1.2 - Frame Section Property Definitions - Summary (Part 2 of 3, continued)

S33Pos in3	S33Neg in3	S22Pos in3	S22Neg in3	Z33 in3	Z22 in3	R33 in	R22 in	Cw in6	Fillet Radius in	CG Offset 3 in	CG Offset 2 in	PNA Offset 3 in
35.39	35.39	12.17	12.17	39.8	18.5	3.5325	2.0486	725.19	0.394	0	0	0
43.29	43.29	15.02	15.02	49	22.9	3.6124	2.0783	929.82	0.395	0	0	0
52.11	52.11	18.27	18.27	59.8	27.9	3.6515	2.0957	1181.76	0.39	0	0	0
60.44	60.44	21.4	21.4	70.1	32.7	3.7158	2.1207	1438.47	0.395	0	0	0

Table 1.2 - Frame Section Property Definitions - Summary (Part 3 of 3)

Area Modifier	As2 Modifier	As3 Modifier	J Modifier	I33 Modifier	I22 Modifier	Mass Modifier	Weight Modifier
1	1	1	0.01	0.75	0.75	1	1
1	1	1	0.01	0.35	0.35	1	1
1	1	1	0.01	0.75	0.75	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1

Table 1.2 - Frame Section Property Definitions - Summary (Part 3 of 3, continued)

Area Modifier	As2 Modifier	As3 Modifier	J Modifier	I33 Modifier	I22 Modifier	Mass Modifier	Weight Modifier
1	1	1	1	1	1	1	1

Table 1.2 - Frame Section Property Definitions - Summary (Part 3 of 3, continued)

Area Modifier	As2 Modifier	As3 Modifier	J Modifier	I33 Modifier	I22 Modifier	Mass Modifier	Weight Modifier
1	1	1	1	1	1	1	1

Table 1.2 - Frame Section Property Definitions - Summary (Part 3 of 3, continued)

Area Modifier	As2 Modifier	As3 Modifier	J Modifier	I33 Modifier	I22 Modifier	Mass Modifier	Weight Modifier
1	1	1	1	1	1	1	1

1.3 Shell Sections

Table 1.3 - Area Section Property Definitions - Summary

Name	Type	Element Type	Material	Total Thickness in	Deck Material	Deck Depth in
Deck1	Deck	Membrane	3000Psi	6.5	A992Fy50	3
Plank1	Slab	Membrane	3000Psi	8		
Slab1	Slab	Shell-Thin	3000Psi	5		
Wall1	Wall	Shell-Thin	3000Psi	12		

1.4 Reinforcement Sizes

Table 1.4 - Reinforcing Bar Sizes

Name	Diameter in	Area in2
#2	0.25	0.05
#3	0.375	0.11
#4	0.5	0.2
#5	0.625	0.31
#6	0.75	0.44
#7	0.875	0.6
#8	1	0.79
#9	1.128	1
#10	1.27	1.27
#11	1.41	1.56
#14	1.693	2.25
#18	2.257	4

1.5 Spring Properties

Table 1.5 - Spring Property Definitions - Area Springs

Name	Stiffness Option	Stiffnes U1 kip/in/in2	Stiffnes U2 kip/in/in2	Stiffnes U3 kip/in/in2	Nonlinear Option for U3	Color	Notes
ASpr1	User	0	0	0.217	Compression Only	Blue	

2 Loads

This chapter provides loading information as applied to the model.

2.1 Load Patterns

Table 2.1 - Load Pattern Definitions

Name	Is Auto Load	Type	Self Weight Multiplier	Auto Load
~LLRF	Yes	Other	0	
Dead	No	Dead	1	
EQ X	No	Seismic	0	ASCE 7-16
EQ Y	No	Seismic	0	ASCE 7-16
Live	No	Live	0	
Soil Pressure	No	Live	0	
Water Pressure	No	Live	0	

2.2 Load Cases

Table 2.2 - Load Case Definitions - Summary

Name	Type
Dead	Linear Static
Live	Linear Static
Modal	Modal - Eigen
Water Pressure	Linear Static
Soil Pressure	Linear Static
EQ X	Linear Static
EQ Y	Linear Static

2.3 Load Combinations

Table 2.3 - Load Combination Definitions

Name	Type	Is Auto	Load Name	SF	Notes
DConS1	Linear Add	Yes	Dead	1.4	Dead [Strength]
DConS2	Linear Add	Yes	Dead	1.2	Dead + Live [Strength]
DConS2			Live	1.6	
DConS2			Water Pressure	1.6	
DConS2			Soil Pressure	1.6	
DConS3	Linear Add	Yes	Dead	1.3	Dead + Live + Static Earthquake [Strength]
DConS3			Live	1	
DConS3			Water Pressure	1	
DConS3			Soil Pressure	1	
DConS3			EQ X	1	
DConS4	Linear Add	Yes	Dead	1.3	Dead + Live - Static Earthquake [Strength]
DConS4			Live	1	
DConS4			Water Pressure	1	
DConS4			Soil Pressure	1	
DConS4			EQ X	-1	
DConS5	Linear Add	Yes	Dead	1.3	Dead + Live + Static Earthquake [Strength]
DConS5			Live	1	

Table 2.3 - Load Combination Definitions (continued)

Name	Type	Is Auto	Load Name	SF	Notes
DConS5			Water Pressure	1	
DConS5			Soil Pressure	1	
DConS5			EQ Y	1	
DConS6	Linear Add	Yes	Dead	1.3	Dead + Live - Static Earthquake [Strength]
DConS6			Live	1	
DConS6			Water Pressure	1	
DConS6			Soil Pressure	1	
DConS6			EQ Y	-1	
DConS7	Linear Add	Yes	Dead	0.8	Dead (min) + Static Earthquake [Strength]
DConS7			EQ X	1	
DConS8	Linear Add	Yes	Dead	0.8	Dead (min) - Static Earthquake [Strength]
DConS8			EQ X	-1	
DConS9	Linear Add	Yes	Dead	0.8	Dead (min) + Static Earthquake [Strength]
DConS9			EQ Y	1	
DConS10	Linear Add	Yes	Dead	0.8	Dead (min) - Static Earthquake [Strength]
DConS10			EQ Y	-1	
DSlbS1	Linear Add	Yes	Dead	1.4	Dead [Strength]
DSlbS2	Linear Add	Yes	Dead	1.2	Dead + Live [Strength]
DSlbS2			Live	1.6	
DSlbS2			Water Pressure	1.6	
DSlbS2			Soil Pressure	1.6	
DSlbS3	Linear Add	Yes	Dead	1.4	Dead [Strength]
DSlbS4	Linear Add	Yes	Dead	1.2	Dead + Live [Strength]
DSlbS4			Live	1.6	
DSlbS4			Water Pressure	1.6	
DSlbS4			Soil Pressure	1.6	
DSlbS5	Linear Add	Yes	Dead	1.3	Dead + Live + Static Earthquake [Strength]
DSlbS5			Live	1	
DSlbS5			Water Pressure	1	
DSlbS5			Soil Pressure	1	
DSlbS5			EQ X	1	
DSlbS6	Linear Add	Yes	Dead	1.3	Dead + Live - Static Earthquake [Strength]
DSlbS6			Live	1	
DSlbS6			Water Pressure	1	
DSlbS6			Soil Pressure	1	
DSlbS6			EQ X	-1	
DSlbS7	Linear Add	Yes	Dead	1.3	Dead + Live + Static Earthquake [Strength]
DSlbS7			Live	1	
DSlbS7			Water Pressure	1	
DSlbS7			Soil Pressure	1	
DSlbS7			EQ Y	1	
DSlbS8	Linear Add	Yes	Dead	1.3	Dead + Live - Static Earthquake [Strength]
DSlbS8			Live	1	
DSlbS8			Water Pressure	1	
DSlbS8			Soil Pressure	1	
DSlbS8			EQ Y	-1	

Table 2.3 - Load Combination Definitions (continued)

Name	Type	Is Auto	Load Name	SF	Notes
DSlbS9	Linear Add	Yes	Dead	0.8	Dead (min) + Static Earthquake [Strength]
DSlbS9			EQ X	1	
DSlbS10	Linear Add	Yes	Dead	0.8	Dead (min) - Static Earthquake [Strength]
DSlbS10			EQ X	-1	
DSlbS11	Linear Add	Yes	Dead	0.8	Dead (min) + Static Earthquake [Strength]
DSlbS11			EQ Y	1	
DSlbS12	Linear Add	Yes	Dead	0.8	Dead (min) - Static Earthquake [Strength]
DSlbS12			EQ Y	-1	
DWalS1	Linear Add	Yes	Dead	1.4	Dead [Strength]
DWalS2	Linear Add	Yes	Dead	1.2	Dead + Live [Strength]
DWalS2			Live	1.6	
DWalS2			Water Pressure	1.6	
DWalS2			Soil Pressure	1.6	
DWalS3	Linear Add	Yes	Dead	1.3	Dead + Live + Static Earthquake [Strength]
DWalS3			Live	1	
DWalS3			Water Pressure	1	
DWalS3			Soil Pressure	1	
DWalS3			EQ X	1	
DWalS4	Linear Add	Yes	Dead	1.3	Dead + Live - Static Earthquake [Strength]
DWalS4			Live	1	
DWalS4			Water Pressure	1	
DWalS4			Soil Pressure	1	
DWalS4			EQ X	-1	
DWalS5	Linear Add	Yes	Dead	1.3	Dead + Live + Static Earthquake [Strength]
DWalS5			Live	1	
DWalS5			Water Pressure	1	
DWalS5			Soil Pressure	1	
DWalS5			EQ Y	1	
DWalS6	Linear Add	Yes	Dead	1.3	Dead + Live - Static Earthquake [Strength]
DWalS6			Live	1	
DWalS6			Water Pressure	1	
DWalS6			Soil Pressure	1	
DWalS6			EQ Y	-1	
DWalS7	Linear Add	Yes	Dead	0.8	Dead (min) + Static Earthquake [Strength]
DWalS7			EQ X	1	
DWalS8	Linear Add	Yes	Dead	0.8	Dead (min) - Static Earthquake [Strength]
DWalS8			EQ X	-1	
DWalS9	Linear Add	Yes	Dead	0.8	Dead (min) + Static Earthquake [Strength]
DWalS9			EQ Y	1	
DWalS10	Linear Add	Yes	Dead	0.8	Dead (min) - Static Earthquake [Strength]
DWalS10			EQ Y	-1	

3 Analysis Results

This chapter provides analysis results.

3.1 Structure Results

Table 3.1 - Base Reactions

Output Case	Case Type	FX kip	FY kip	FZ kip	MX kip-ft	MY kip-ft	MZ kip-ft	X ft	Y ft	Z ft
Dead	LinStatic	0	0	40330.803	5404849.2597	-3503333	0	0	0	0
Live	LinStatic	0	0	1539.378	206752.7143	-133733.4908	0	0	0	0
Water Pressure	LinStatic	0	0	94.478	22229.1386	-8207.7328	0	0	0	0
Soil Pressure	LinStatic	0	0	-52.992	-12468.2668	4603.6963	0	0	0	0
EQ X	LinStatic	-1873.716	0	0	0	-69938.6634	250707.1984	0	0	0
EQ Y	LinStatic	0	-1873.716	0	69938.6634	0	-162765.9823	0	0	0
DSlbS1	Combination	0	0	56463.125	7566788.9635	-4904666	0	0	0	0
DSlbS2	Combination	0	0	50926.346	6832240.8493	-4423739	0	0	0	0
DSlbS3	Combination	0	0	56463.125	7566788.9635	-4904666	0	0	0	0
DSlbS4	Combination	0	0	50926.346	6832240.8493	-4423739	0	0	0	0
DSlbS5	Combination	-1873.716	0	54010.908	7242817.6236	-4761609	250707.1984	0	0	0
DSlbS6	Combination	1873.716	0	54010.908	7242817.6236	-4621731	-250707.1984	0	0	0
DSlbS7	Combination	0	-1873.716	54010.908	7312756.287	-4691670	-162765.9823	0	0	0
DSlbS8	Combination	0	1873.716	54010.908	7172878.9602	-4691670	162765.9823	0	0	0
DSlbS9	Combination	-1873.716	0	32264.643	4323879.4077	-2872605	250707.1984	0	0	0
DSlbS10	Combination	1873.716	0	32264.643	4323879.4077	-2732728	-250707.1984	0	0	0
DSlbS11	Combination	0	-1873.716	32264.643	4393818.0711	-2802666	-162765.9823	0	0	0
DSlbS12	Combination	0	1873.716	32264.643	4253940.7443	-2802666	162765.9823	0	0	0
DConS1	Combination	0	0	56463.125	7566788.9635	-4904666	0	0	0	0
DConS2	Combination	0	0	50926.346	6832240.8493	-4423739	0	0	0	0
DConS3	Combination	-1873.716	0	54010.908	7242817.6236	-4761609	250707.1984	0	0	0

Table 3.1 - Base Reactions (continued)

Output Case	Case Type	FX kip	FY kip	FZ kip	MX kip-ft	MY kip-ft	MZ kip-ft	X ft	Y ft	Z ft
DConS4	Combination	1873.716	0	54010.908	7242817.6236	-4621731	-250707.1984	0	0	0
DConS5	Combination	0	1873.716	54010.908	7312756.287	-4691670	-162765.9823	0	0	0
DConS6	Combination	0	1873.716	54010.908	7172878.9602	-4691670	162765.9823	0	0	0
DConS7	Combination	-1873.716	0	32264.643	4323879.4077	-2872605	250707.1984	0	0	0
DConS8	Combination	1873.716	0	32264.643	4323879.4077	-2732728	-250707.1984	0	0	0
DConS9	Combination	0	1873.716	32264.643	4393818.0711	-2802666	-162765.9823	0	0	0
DConS10	Combination	0	1873.716	32264.643	4253940.7443	-2802666	162765.9823	0	0	0
DWalS1	Combination	0	0	56463.125	7566788.9635	-4904666	0	0	0	0
DWalS2	Combination	0	0	50926.346	6832240.8493	-4423739	0	0	0	0
DWalS3	Combination	-1873.716	0	54010.908	7242817.6236	-4761609	250707.1984	0	0	0
DWalS4	Combination	1873.716	0	54010.908	7242817.6236	-4621731	-250707.1984	0	0	0
DWalS5	Combination	0	1873.716	54010.908	7312756.287	-4691670	-162765.9823	0	0	0
DWalS6	Combination	0	1873.716	54010.908	7172878.9602	-4691670	162765.9823	0	0	0
DWalS7	Combination	-1873.716	0	32264.643	4323879.4077	-2872605	250707.1984	0	0	0
DWalS8	Combination	1873.716	0	32264.643	4323879.4077	-2732728	-250707.1984	0	0	0
DWalS9	Combination	0	1873.716	32264.643	4393818.0711	-2802666	-162765.9823	0	0	0
DWalS10	Combination	0	1873.716	32264.643	4253940.7443	-2802666	162765.9823	0	0	0

3.2 Story Results

Table 3.2 - Story Drifts

Story	Output Case	Case Type	Direction	Drift	Label	X ft	Y ft	Z ft
F4	Dead	LinStatic	Y	1E-05	30	0	258.743	52
F4	Live	LinStatic	Y	1E-06	99	76.25	228.16	52
F4	Water Pressure	LinStatic	Y	1.052E-07	104	82.25	242.41	52
F4	Soil Pressure	LinStatic	Y	5.87E-08	104	82.25	242.41	52
F4	EQ X	LinStatic	X	0.000206	156	144.5	199.66	52
F4	EQ Y	LinStatic	Y	0.000209	106	82.25	258.743	52
F4	DSIbS1	Combination	Y	1.4E-05	30	0	258.743	52

Table 3.2 - Story Drifts (continued)

Story	Output Case	Case Type	Direction	Drift	Label	X ft	Y ft	Z ft
F4	DSlbS2	Combination	Y	1.2E-05	30	0	258.743	52
F4	DSlbS3	Combination	Y	1.4E-05	30	0	258.743	52
F4	DSlbS4	Combination	Y	1.2E-05	30	0	258.743	52
F4	DSlbS5	Combination	X	0.000212	154	144.5	171.16	52
F4	DSlbS6	Combination	X	0.000211	81	29.25	171.16	52
F4	DSlbS7	Combination	Y	0.000201	103	82.25	228.16	52
F4	DSlbS8	Combination	Y	0.00022	106	82.25	258.743	52
F4	DSlbS9	Combination	X	0.000209	154	144.5	171.16	52
F4	DSlbS10	Combination	X	0.000209	82	29.25	185.41	52
F4	DSlbS11	Combination	Y	0.000205	103	82.25	228.16	52
F4	DSlbS12	Combination	Y	0.000215	106	82.25	258.743	52
F4	DConS1	Combination	Y	1.4E-05	30	0	258.743	52
F4	DConS2	Combination	Y	1.2E-05	30	0	258.743	52
F4	DConS3	Combination	X	0.000212	154	144.5	171.16	52
F4	DConS4	Combination	X	0.000211	81	29.25	171.16	52
F4	DConS5	Combination	Y	0.000201	103	82.25	228.16	52
F4	DConS6	Combination	Y	0.00022	106	82.25	258.743	52
F4	DConS7	Combination	X	0.000209	154	144.5	171.16	52
F4	DConS8	Combination	X	0.000209	82	29.25	185.41	52
F4	DConS9	Combination	Y	0.000205	103	82.25	228.16	52
F4	DConS10	Combination	Y	0.000215	106	82.25	258.743	52
F4	DWalS1	Combination	Y	1.4E-05	30	0	258.743	52
F4	DWalS2	Combination	Y	1.2E-05	30	0	258.743	52
F4	DWalS3	Combination	X	0.000212	154	144.5	171.16	52
F4	DWalS4	Combination	X	0.000211	81	29.25	171.16	52
F4	DWalS5	Combination	Y	0.000201	103	82.25	228.16	52
F4	DWalS6	Combination	Y	0.00022	106	82.25	258.743	52
F4	DWalS7	Combination	X	0.000209	154	144.5	171.16	52
F4	DWalS8	Combination	X	0.000209	82	29.25	185.41	52
F4	DWalS9	Combination	Y	0.000205	103	82.25	228.16	52
F4	DWalS10	Combination	Y	0.000215	106	82.25	258.743	52
F3	Dead	LinStatic	Y	7E-06	111	97.5	228.16	41
F3	Live	LinStatic	Y	4.721E-07	137	164.5	258.743	41
F3	Water Pressure	LinStatic	Y	4.974E-08	106	82.25	258.743	41
F3	Soil Pressure	LinStatic	Y	2.749E-08	103	82.25	228.16	41
F3	EQ X	LinStatic	X	0.000347	115	111.75	228.16	41
F3	EQ Y	LinStatic	Y	0.000361	105	82.25	249.493	41
F3	DSlbS1	Combination	Y	9E-06	111	97.5	228.16	41
F3	DSlbS2	Combination	Y	8E-06	111	97.5	228.16	41
F3	DSlbS3	Combination	Y	9E-06	111	97.5	228.16	41
F3	DSlbS4	Combination	Y	8E-06	111	97.5	228.16	41
F3	DSlbS5	Combination	X	0.000348	49	9.25	258.743	41
F3	DSlbS6	Combination	X	0.000347	115	111.75	228.16	41
F3	DSlbS7	Combination	Y	0.000353	105	82.25	249.493	41
F3	DSlbS8	Combination	Y	0.000369	105	82.25	249.493	41

Table 3.2 - Story Drifts (continued)

Story	Output Case	Case Type	Direction	Drift	Label	X ftft	Y	Z ft
F3	DSlbS9	Combination	X	0.000347	49	9.25	258.743	41
F3	DSlbS10	Combination	X	0.000347	111	97.5	228.16	41
F3	DSlbS11	Combination	Y	0.000356	105	82.25	249.493	41
F3	DSlbS12	Combination	Y	0.000366	105	82.25	249.493	41
F3	DConS1	Combination	Y	9E-06	111	97.5	228.16	41
F3	DConS2	Combination	Y	8E-06	111	97.5	228.16	41
F3	DConS3	Combination	X	0.000348	49	9.25	258.743	41
F3	DConS4	Combination	X	0.000347	115	111.75	228.16	41
F3	DConS5	Combination	Y	0.000353	105	82.25	249.493	41
F3	DConS6	Combination	Y	0.000369	105	82.25	249.493	41
F3	DConS7	Combination	X	0.000347	49	9.25	258.743	41
F3	DConS8	Combination	X	0.000347	111	97.5	228.16	41
F3	DConS9	Combination	Y	0.000356	105	82.25	249.493	41
F3	DConS10	Combination	Y	0.000366	105	82.25	249.493	41
F3	DWalS1	Combination	Y	9E-06	111	97.5	228.16	41
F3	DWalS2	Combination	Y	8E-06	111	97.5	228.16	41
F3	DWalS3	Combination	X	0.000348	49	9.25	258.743	41
F3	DWalS4	Combination	X	0.000347	115	111.75	228.16	41
F3	DWalS5	Combination	Y	0.000353	105	82.25	249.493	41
F3	DWalS6	Combination	Y	0.000369	105	82.25	249.493	41
F3	DWalS7	Combination	X	0.000347	49	9.25	258.743	41
F3	DWalS8	Combination	X	0.000347	111	97.5	228.16	41
F3	DWalS9	Combination	Y	0.000356	105	82.25	249.493	41
F3	DWalS10	Combination	Y	0.000366	105	82.25	249.493	41
F2	Dead	LinStatic	X	2E-06	30	0	258.743	30
F2	Dead	LinStatic	Y	3E-06	137	164.5	258.743	30
F2	Live	LinStatic	X	1.275E-07	137	164.5	258.743	30
F2	Live	LinStatic	Y	3.116E-07	49	9.25	258.743	30
F2	Water Pressure	LinStatic	Y	6.801E-08	99	76.25	228.16	30
F2	Soil Pressure	LinStatic	Y	2.989E-08	99	76.25	228.16	30
F2	EQ X	LinStatic	X	0.000475	107	91.5	228.16	30
F2	EQ Y	LinStatic	Y	0.000523	103	82.25	228.16	30
F2	DSlbS1	Combination	X	2E-06	30	0	258.743	30
F2	DSlbS1	Combination	Y	4E-06	137	164.5	258.743	30
F2	DSlbS2	Combination	X	2E-06	30	0	258.743	30
F2	DSlbS2	Combination	Y	4E-06	137	164.5	258.743	30
F2	DSlbS3	Combination	X	2E-06	30	0	258.743	30
F2	DSlbS3	Combination	Y	4E-06	137	164.5	258.743	30
F2	DSlbS4	Combination	X	2E-06	30	0	258.743	30
F2	DSlbS4	Combination	Y	4E-06	137	164.5	258.743	30
F2	DSlbS5	Combination	X	0.000475	103	82.25	228.16	30
F2	DSlbS6	Combination	X	0.000475	107	91.5	228.16	30
F2	DSlbS7	Combination	Y	0.000521	103	82.25	228.16	30
F2	DSlbS8	Combination	Y	0.000525	103	82.25	228.16	30
F2	DSlbS9	Combination	X	0.000475	103	82.25	228.16	30

Table 3.2 - Story Drifts (continued)

Story	Output Case	Case Type	Direction	Drift	Label	X ftft	Y	Z ft
F2	DSlbS10	Combination	X	0.000475	107	91.5	228.16	30
F2	DSlbS11	Combination	Y	0.000521	103	82.25	228.16	30
F2	DSlbS12	Combination	Y	0.000524	103	82.25	228.16	30
F2	DConS1	Combination	X	2E-06	30	0	258.743	30
F2	DConS1	Combination	Y	4E-06	137	164.5	258.743	30
F2	DConS2	Combination	X	2E-06	30	0	258.743	30
F2	DConS2	Combination	Y	4E-06	137	164.5	258.743	30
F2	DConS3	Combination	X	0.000475	103	82.25	228.16	30
F2	DConS4	Combination	X	0.000475	107	91.5	228.16	30
F2	DConS5	Combination	Y	0.000521	103	82.25	228.16	30
F2	DConS6	Combination	Y	0.000525	103	82.25	228.16	30
F2	DConS7	Combination	X	0.000475	103	82.25	228.16	30
F2	DConS8	Combination	X	0.000475	107	91.5	228.16	30
F2	DConS9	Combination	Y	0.000521	103	82.25	228.16	30
F2	DConS10	Combination	Y	0.000524	103	82.25	228.16	30
F2	DWalS1	Combination	X	2E-06	30	0	258.743	30
F2	DWalS1	Combination	Y	4E-06	137	164.5	258.743	30
F2	DWalS2	Combination	X	2E-06	30	0	258.743	30
F2	DWalS2	Combination	Y	4E-06	137	164.5	258.743	30
F2	DWalS3	Combination	X	0.000475	103	82.25	228.16	30
F2	DWalS4	Combination	X	0.000475	107	91.5	228.16	30
F2	DWalS5	Combination	Y	0.000521	103	82.25	228.16	30
F2	DWalS6	Combination	Y	0.000525	103	82.25	228.16	30
F2	DWalS7	Combination	X	0.000475	103	82.25	228.16	30
F2	DWalS8	Combination	X	0.000475	107	91.5	228.16	30
F2	DWalS9	Combination	Y	0.000521	103	82.25	228.16	30
F2	DWalS10	Combination	Y	0.000524	103	82.25	228.16	30
F1	Dead	LinStatic	X	3E-06	140	173.75	249.493	19
F1	Dead	LinStatic	Y	3E-06	141	173.75	258.743	19
F1	Live	LinStatic	X	2.934E-07	144	144.5	28.66	19
F1	Live	LinStatic	Y	3.159E-07	111	97.5	228.16	19
F1	Water Pressure	LinStatic	Y	4.41E-07	99	76.25	228.16	19
F1	Soil Pressure	LinStatic	Y	2.367E-07	99	76.25	228.16	19
F1	EQ X	LinStatic	X	0.000569	103	82.25	228.16	19
F1	EQ Y	LinStatic	Y	0.00062	99	76.25	228.16	19
F1	DSlbS1	Combination	X	4E-06	140	173.75	249.493	19
F1	DSlbS1	Combination	Y	5E-06	141	173.75	258.743	19
F1	DSlbS2	Combination	X	3E-06	140	173.75	249.493	19
F1	DSlbS2	Combination	Y	4E-06	141	173.75	258.743	19
F1	DSlbS3	Combination	X	4E-06	140	173.75	249.493	19
F1	DSlbS3	Combination	Y	5E-06	141	173.75	258.743	19
F1	DSlbS4	Combination	X	3E-06	140	173.75	249.493	19
F1	DSlbS4	Combination	Y	4E-06	141	173.75	258.743	19
F1	DSlbS5	Combination	X	0.00057	103	82.25	228.16	19
F1	DSlbS6	Combination	X	0.000569	107	91.5	228.16	19

Table 3.2 - Story Drifts (continued)

Story	Output Case	Case Type	Direction	Drift	Label	X ftft	Y	Z ft
F1	DSlbS7	Combination	Y	0.00062	99	76.25	228.16	19
F1	DSlbS8	Combination	Y	0.00062	111	97.5	228.16	19
F1	DSlbS9	Combination	X	0.000569	103	82.25	228.16	19
F1	DSlbS10	Combination	X	0.000569	107	91.5	228.16	19
F1	DSlbS11	Combination	Y	0.00062	99	76.25	228.16	19
F1	DSlbS12	Combination	Y	0.000619	111	97.5	228.16	19
F1	DConS1	Combination	X	4E-06	140	173.75	249.493	19
F1	DConS1	Combination	Y	5E-06	141	173.75	258.743	19
F1	DConS2	Combination	X	3E-06	140	173.75	249.493	19
F1	DConS2	Combination	Y	4E-06	141	173.75	258.743	19
F1	DConS3	Combination	X	0.00057	103	82.25	228.16	19
F1	DConS4	Combination	X	0.000569	107	91.5	228.16	19
F1	DConS5	Combination	Y	0.00062	99	76.25	228.16	19
F1	DConS6	Combination	Y	0.00062	111	97.5	228.16	19
F1	DConS7	Combination	X	0.000569	103	82.25	228.16	19
F1	DConS8	Combination	X	0.000569	107	91.5	228.16	19
F1	DConS9	Combination	Y	0.00062	99	76.25	228.16	19
F1	DConS10	Combination	Y	0.000619	111	97.5	228.16	19
F1	DWalS1	Combination	X	4E-06	140	173.75	249.493	19
F1	DWalS1	Combination	Y	5E-06	141	173.75	258.743	19
F1	DWalS2	Combination	X	3E-06	140	173.75	249.493	19
F1	DWalS2	Combination	Y	4E-06	141	173.75	258.743	19
F1	DWalS3	Combination	X	0.00057	103	82.25	228.16	19
F1	DWalS4	Combination	X	0.000569	107	91.5	228.16	19
F1	DWalS5	Combination	Y	0.00062	99	76.25	228.16	19
F1	DWalS6	Combination	Y	0.00062	111	97.5	228.16	19
F1	DWalS7	Combination	X	0.000569	103	82.25	228.16	19
F1	DWalS8	Combination	X	0.000569	107	91.5	228.16	19
F1	DWalS9	Combination	Y	0.00062	99	76.25	228.16	19
F1	DWalS10	Combination	Y	0.000619	111	97.5	228.16	19
Plinth Level	Dead	LinStatic	Y	1.5E-05	91	47.75	228.16	7
Plinth Level	Live	LinStatic	Y	4E-06	91	47.75	228.16	7
Plinth Level	Water Pressure	LinStatic	Y	2E-06	99	76.25	228.16	7
Plinth Level	Soil Pressure	LinStatic	Y	1E-06	99	76.25	228.16	7
Plinth Level	EQ X	LinStatic	X	0.000595	76	29.25	99.91	7
Plinth Level	EQ Y	LinStatic	Y	0.000653	141	173.75	258.743	7
Plinth Level	DSlbS1	Combination	Y	2.1E-05	91	47.75	228.16	7
Plinth Level	DSlbS2	Combination	Y	2.4E-05	91	47.75	228.16	7
Plinth Level	DSlbS3	Combination	Y	2.1E-05	91	47.75	228.16	7
Plinth Level	DSlbS4	Combination	Y	2.4E-05	91	47.75	228.16	7
Plinth Level	DSlbS5	Combination	X	0.000604	18	0	99.91	7
Plinth Level	DSlbS6	Combination	X	0.000601	200	173.75	99.91	7
Plinth Level	DSlbS7	Combination	Y	0.000653	141	173.75	258.743	7
Plinth Level	DSlbS8	Combination	Y	0.000653	141	173.75	258.743	7
Plinth Level	DSlbS9	Combination	X	0.0006	18	0	99.91	7

Table 3.2 - Story Drifts (continued)

Story	Output Case	Case Type	Direction	Drift	Label	X ftft	Y	Z ft
Plinth Level Plinth	DSlbS10	Combination	X	0.000598	76	29.25	99.91	7
Level Plinth Level	DSlbS11	Combination	Y	0.000653	141	173.75	258.743	7
Plinth Level Plinth	DSlbS12	Combination	Y	0.000653	141	173.75	258.743	7
Level Plinth Level	DConS1	Combination	Y	2.1E-05	91	47.75	228.16	7
Plinth Level Plinth	DConS2	Combination	Y	2.4E-05	91	47.75	228.16	7
Level Plinth Level	DConS3	Combination	X	0.000604	18	0	99.91	7
Plinth Level Plinth	DConS4	Combination	X	0.000601	200	173.75	99.91	7
Level Plinth Level	DConS5	Combination	Y	0.000653	141	173.75	258.743	7
Plinth Level Plinth	DConS6	Combination	Y	0.000653	141	173.75	258.743	7
Level Plinth Level	DConS7	Combination	X	0.0006	18	0	99.91	7
Plinth Level Plinth	DConS8	Combination	X	0.000598	76	29.25	99.91	7
Level Plinth Level	DConS9	Combination	Y	0.000653	141	173.75	258.743	7
Plinth Level Plinth	DConS10	Combination	Y	0.000653	141	173.75	258.743	7
Level Plinth Level	DWalS1	Combination	Y	2.1E-05	91	47.75	228.16	7
Plinth Level Plinth	DWalS2	Combination	Y	2.4E-05	91	47.75	228.16	7
Level Under Water	DWalS3	Combination	X	0.000604	18	0	99.91	7
Level Under Water	DWalS4	Combination	X	0.000601	200	173.75	99.91	7
Level Under Water	DWalS5	Combination	Y	0.000653	141	173.75	258.743	7
Level Under Water	DWalS6	Combination	Y	0.000653	141	173.75	258.743	7
Level Under Water	DWalS7	Combination	X	0.0006	18	0	99.91	7
Level Under Water	DWalS8	Combination	X	0.000598	76	29.25	99.91	7
Level Under Water	DWalS9	Combination	Y	0.000653	141	173.75	258.743	7
Level Under Water	DWalS10	Combination	Y	0.000653	141	173.75	258.743	7
Level Under Water	Dead	LinStatic	X	2E-05	229	97.5	28.66	2
Level Under Water	Dead	LinStatic	Y	2.9E-05	119	126	228.16	2
Level Under Water	Live	LinStatic	Y	8E-06	119	126	228.16	2
Level Under Water	Water Pressure	LinStatic	Y	4E-06	100	76.25	242.41	2
Level Under Water	Soil Pressure	LinStatic	Y	2E-06	100	76.25	242.41	2
Level Under Water	EQ X	LinStatic	X	0.000751	226	91.5	28.66	2
Level Under Water	EQ Y	LinStatic	Y	0.000829	140	173.75	249.493	2
Level Under Water	DSlbS1	Combination	X	2.8E-05	229	97.5	28.66	2
Level Under Water	DSlbS1	Combination	Y	4E-05	119	126	228.16	2
Level Under Water	DSlbS2	Combination	X	3E-05	229	97.5	28.66	2
Level Under Water	DSlbS2	Combination	Y	4.8E-05	119	126	228.16	2
Level Under Water	DSlbS3	Combination	X	2.8E-05	229	97.5	28.66	2
Level Under Water	DSlbS3	Combination	Y	4E-05	119	126	228.16	2
Level Under Water	DSlbS4	Combination	X	3E-05	229	97.5	28.66	2
Level	DSlbS4	Combination	Y	4.8E-05	119	126	228.16	2
	DSlbS5	Combination	X	0.000759	226	91.5	28.66	2
	DSlbS6	Combination	X	0.000754	166	153.75	99.91	2
	DSlbS7	Combination	Y	0.000839	139	173.75	242.41	2
	DSlbS8	Combination	Y	0.000826	29	0	249.493	2
	DSlbS9	Combination	X	0.000755	226	91.5	28.66	2
	DSlbS10	Combination	X	0.000751	166	153.75	99.91	2
	DSlbS11	Combination	Y	0.000833	140	173.75	249.493	2

Table 3.2 - Story Drifts (continued)

Story	Output Case	Case Type	Direction	Drift	Label	X ftft	Y	Z ft
Under Water Level	DSlbS12	Combination	Y	0.000825	140	173.75		2
Under Water Level	DConS1	Combination	X	2.8E-05	229	249.493		2
Under Water Level	DConS1	Combination	Y	4E-05	119	97.5	28.66	2
Under Water Level	DConS2	Combination	X	3E-05	229	126	228.16	2
Under Water Level	DConS2	Combination	Y	4.8E-05	119	97.5	28.66	2
Under Water Level	DConS3	Combination	X	0.000759	226	126	228.16	2
Under Water Level	DConS4	Combination	X	0.000754	166	91.5	28.66	2
Under Water Level	DConS5	Combination	Y	0.000839	139	153.75	99.91	2
Under Water Level	DConS6	Combination	Y	0.000826	29	173.75	242.41	2
Under Water Level	DConS7	Combination	X	0.000755	226	0	249.493	2
Under Water Level	DConS8	Combination	X	0.000751	166	91.5	28.66	2
Under Water Level	DConS9	Combination	Y	0.000833	140	153.75	99.91	2
Under Water Level	DConS10	Combination	Y	0.000825	140	173.75	249.493	2
Under Water Level	DWalS1	Combination	X	2.8E-05	229	173.75	249.493	2
Under Water Level	DWalS1	Combination	Y	4E-05	119	97.5	28.66	2
Under Water Level	DWalS2	Combination	X	3E-05	229	126	228.16	2
Under Water Level	DWalS2	Combination	Y	4.8E-05	119	97.5	28.66	2
Under Water Level	DWalS3	Combination	X	0.000759	226	126	228.16	2
Under Water Level	DWalS4	Combination	X	0.000754	166	91.5	28.66	2
Under Water Level	DWalS5	Combination	Y	0.000839	139	153.75	99.91	2
Under Water Level	DWalS6	Combination	Y	0.000826	29	173.75	242.41	2
Under Water Level	DWalS7	Combination	X	0.000755	226	0	249.493	2
Under Water Level	DWalS8	Combination	X	0.000751	166	91.5	28.66	2
Under Water Level	DWalS9	Combination	Y	0.000833	140	153.75	99.91	2
Under Water Level	DWalS10	Combination	Y	0.000825	140	173.75	249.493	2
						173.75	249.493	

Table 3.3 - Story Forces

Story	Output Case	Case Type	Location	P kip	VX kip	VY kip	T kip-ft	MX kip-ft	MY kip-ft
F4	Dead	LinStatic	Top	4525.709	0	0	0	607036.9081	-393171.0039
F4	Dead	LinStatic	Bottom	6114.464	0	0	0	820502.9109	-531194.0353
F4	Live	LinStatic	Top	0	0	0	0	0	0
F4	Live	LinStatic	Bottom	1.771	0	0	0	56.2986	-153.8757
F4	Water Pressure	LinStatic	Top	0	0	0	0	0	0
F4	Water Pressure	LinStatic	Bottom	0	0	0	0	0	0
F4	Soil Pressure	LinStatic	Top	0	0	0	0	0	0
F4	Soil Pressure	LinStatic	Bottom	0	0	0	0	0	0
F4	EQ X	LinStatic	Top	0	-578.854	0	77450.4472	0	0
F4	EQ X	LinStatic	Bottom	0	-580.417	0	77729.2559	0	-6375.9941
F4	EQ Y	LinStatic	Top	0	0	-578.854	-50288.4467	0	0
F4	EQ Y	LinStatic	Bottom	0	0	-580.417	-50424.0806	6375.99410	

Table 3.3 - Story Forces (continued)

Story	Output Case	Case Type	Location	P kip	VX kip	VY kip	T kip-ft	MX kip-ft	MY kip-ft
F4	DSIbS1	Combination	Top	6335.993	0	0	0	849851.6713	-550439.4054
F4	DSIbS1	Combination	Bottom	8560.249	0	0	0	1148704.0753	-743671.6495
F4	DSIbS2	Combination	Top	5430.851	0	0	0	728444.2897	-471805.2046
F4	DSIbS2	Combination	Bottom	7340.19	0	0	0	984693.5708	-637679.0436
F4	DSIbS3	Combination	Top	6335.993	0	0	0	849851.6713	-550439.4054
F4	DSIbS3	Combination	Bottom	8560.249	0	0	0	1148704.0753	-743671.6495
F4	DSIbS4	Combination	Top	5430.851	0	0	0	728444.2897	-471805.2046
F4	DSIbS4	Combination	Bottom	7340.19	0	0	0	984693.5708	-637679.0436
F4	DSIbS5	Combination	Top	5883.422	-578.854	0	77450.4472	789147.9805	-511122.305
F4	DSIbS5	Combination	Bottom	7950.574	-580.417	0	77729.2559	1066710.0828	-697082.1158
F4	DSIbS6	Combination	Top	5883.422	578.854	0	-77450.4472	789147.9805	-511122.305
F4	DSIbS6	Combination	Bottom	7950.574	580.417	0	-77729.2559	1066710.0828	-684330.1276
F4	DSIbS7	Combination	Top	5883.422	0	-578.854	-50288.4467	789147.9805	-511122.305
F4	DSIbS7	Combination	Bottom	7950.574	0	-580.417	-50424.0806	1073086.0769	-690706.1217
F4	DSIbS8	Combination	Top	5883.422	0	578.854	50288.4467	789147.9805	-511122.305
F4	DSIbS8	Combination	Bottom	7950.574	0	580.417	50424.0806	1060334.0887	-690706.1217
F4	DSIbS9	Combination	Top	3620.568	-578.854	0	77450.4472	485629.5265	-314536.8031
F4	DSIbS9	Combination	Bottom	4891.571	-580.417	0	77729.2559	656402.3287	-431331.2224
F4	DSIbS10	Combination	Top	3620.568	578.854	0	-77450.4472	485629.5265	-314536.8031
F4	DSIbS10	Combination	Bottom	4891.571	580.417	0	-77729.2559	656402.3287	-418579.2342
F4	DSIbS11	Combination	Top	3620.568	0	-578.854	-50288.4467	485629.5265	-314536.8031
F4	DSIbS11	Combination	Bottom	4891.571	0	-580.417	-50424.0806	662778.3228	-424955.2283
F4	DSIbS12	Combination	Top	3620.568	0	578.854	50288.4467	485629.5265	-314536.8031

Table 3.3 - Story Forces (continued)

Story	Output Case	Case Type	Location	P kip	VX kip	VY kip	T kip-ft	MX kip-ft	MY kip-ft
F4	DSlbS12	Combination	Bottom	4891.571	0	580.417	50424.0806	650026.3346	-424955.2283
F4	DConS1	Combination	Top	6335.993	0	0	0	849851.6713	-550439.4054
F4	DConS1	Combination	Bottom	8560.249	0	0	0	1148704.0753	-743671.6495
F4	DConS2	Combination	Top	5430.851	0	0	0	728444.2897	-471805.2046
F4	DConS2	Combination	Bottom	7340.19	0	0	0	984693.5708	-637679.0436
F4	DConS3	Combination	Top	5883.422	-578.854	0	77450.4472	789147.9805	-511122.305
F4	DConS3	Combination	Bottom	7950.574	-580.417	0	77729.2559	1066710.0828	-697082.1158
F4	DConS4	Combination	Top	5883.422	578.854	0	-77450.4472	789147.9805	-511122.305
F4	DConS4	Combination	Bottom	7950.574	580.417	0	-77729.2559	1066710.0828	-684330.1276
F4	DConS5	Combination	Top	5883.422	0	-578.854	-50288.4467	789147.9805	-511122.305
F4	DConS5	Combination	Bottom	7950.574	0	-580.417	-50424.0806	1073086.0769	-690706.1217
F4	DConS6	Combination	Top	5883.422	0	578.854	50288.4467	789147.9805	-511122.305
F4	DConS6	Combination	Bottom	7950.574	0	580.417	50424.0806	1060334.0887	-690706.1217
F4	DConS7	Combination	Top	3620.568	-578.854	0	77450.4472	485629.5265	-314536.8031
F4	DConS7	Combination	Bottom	4891.571	-580.417	0	77729.2559	656402.3287	-431331.2224
F4	DConS8	Combination	Top	3620.568	578.854	0	-77450.4472	485629.5265	-314536.8031
F4	DConS8	Combination	Bottom	4891.571	580.417	0	-77729.2559	656402.3287	-418579.2342
F4	DConS9	Combination	Top	3620.568	0	-578.854	-50288.4467	485629.5265	-314536.8031
F4	DConS9	Combination	Bottom	4891.571	0	-580.417	-50424.0806	662778.3228	-424955.2283
F4	DConS10	Combination	Top	3620.568	0	578.854	50288.4467	485629.5265	-314536.8031
F4	DConS10	Combination	Bottom	4891.571	0	580.417	50424.0806	650026.3346	-424955.2283
F4	DWalS1	Combination	Top	6335.993	0	0	0	849851.6713	-550439.4054
F4	DWalS1	Combination	Bottom	8560.249	0	0	0	1148704.0753	-743671.6495

Table 3.3 - Story Forces (continued)

Story	Output Case	Case Type	Location	P kip	VX kip	VY kip	T kip-ft	MX kip-ft	MY kip-ft
F4	DWalS2	Combination	Top	5430.851	0	0	0	728444.2897	-471805.2046
F4	DWalS2	Combination	Bottom	7340.19	0	0	0	984693.5708	-637679.0436
F4	DWalS3	Combination	Top	5883.422	-578.854	0	77450.4472	789147.9805	-511122.305
F4	DWalS3	Combination	Bottom	7950.574	-580.417	0	77729.2559	1066710.0828	-697082.1158
F4	DWalS4	Combination	Top	5883.422	578.854	0	-77450.4472	789147.9805	-511122.305
F4	DWalS4	Combination	Bottom	7950.574	580.417	0	-77729.2559	1066710.0828	-684330.1276
F4	DWalS5	Combination	Top	5883.422	0	-578.854	-50288.4467	789147.9805	-511122.305
F4	DWalS5	Combination	Bottom	7950.574	0	-580.417	-50424.0806	1073086.0769	-690706.1217
F4	DWalS6	Combination	Top	5883.422	0	578.854	50288.4467	789147.9805	-511122.305
F4	DWalS6	Combination	Bottom	7950.574	0	580.417	50424.0806	1060334.0887	-690706.1217
F4	DWalS7	Combination	Top	3620.568	-578.854	0	77450.4472	485629.5265	-314536.8031
F4	DWalS7	Combination	Bottom	4891.571	-580.417	0	77729.2559	656402.3287	-431331.2224
F4	DWalS8	Combination	Top	3620.568	578.854	0	-77450.4472	485629.5265	-314536.8031
F4	DWalS8	Combination	Bottom	4891.571	580.417	0	-77729.2559	656402.3287	-418579.2342
F4	DWalS9	Combination	Top	3620.568	0	-578.854	-50288.4467	485629.5265	-314536.8031
F4	DWalS9	Combination	Bottom	4891.571	0	-580.417	-50424.0806	662778.3228	-424955.2283
F4	DWalS10	Combination	Top	3620.568	0	578.854	50288.4467	485629.5265	-314536.8031
F4	DWalS10	Combination	Bottom	4891.571	0	580.417	50424.0806	650026.3346	-424955.2283
F3	Dead	LinStatic	Top	13301.531	0	0	0	1782302.3061	-1155571
F3	Dead	LinStatic	Bottom	14897.223	0	0	0	1996031.3095	-1294196
F3	Live	LinStatic	Top	1.771	0	0	0	56.2986	-153.8757
F3	Live	LinStatic	Bottom	3.542	0	0	0	112.5972	-307.7515
F3	Water Pressure	LinStatic	Top	0	0	0	0	0	0
F3	Water Pressure	LinStatic	Bottom	0	0	0	0	0	0

Table 3.3 - Story Forces (continued)

Story	Output Case	Case Type	Location	P kip	VX kip	VY kip	T kip-ft	MX kip-ft	MY kip-ft
F3	Soil Pressure	LinStatic	Top	0	0	0	0	0	0
F3	Soil Pressure	LinStatic	Bottom	0	0	0	0	0	0
F3	EQ X	LinStatic	Top	0	-1117.815	0	149516.1537	0	-6375.9941
F3	EQ X	LinStatic	Bottom	0	-1119.048	0	149736.2964	0	-18678.7423
F3	EQ Y	LinStatic	Top	0	0	-1117.815	-97110.5565	6375.9941	0
F3	EQ Y	LinStatic	Bottom	0	0	-1119.048	-97217.7165	18678.7423	0
F3	DSlbS1	Combination	Top	18622.144	0	0	0	2495223.2285	-1617799
F3	DSlbS1	Combination	Bottom	20856.113	0	0	0	2794443.8333	-1811875
F3	DSlbS2	Combination	Top	15964.672	0	0	0	2138852.845	-1386931
F3	DSlbS2	Combination	Bottom	17882.336	0	0	0	2395417.7269	-1553528
F3	DSlbS3	Combination	Top	18622.144	0	0	0	2495223.2285	-1617799
F3	DSlbS3	Combination	Bottom	20856.113	0	0	0	2794443.8333	-1811875
F3	DSlbS4	Combination	Top	15964.672	0	0	0	2138852.845	-1386931
F3	DSlbS4	Combination	Bottom	17882.336	0	0	0	2395417.7269	-1553528
F3	DSlbS5	Combination	Top	17293.762	-1117.815	0	149516.1537	2317049.2965	-1508772
F3	DSlbS5	Combination	Bottom	19369.933	-1119.048	0	149736.2964	2594953.2995	-1701442
F3	DSlbS6	Combination	Top	17293.762	1117.815	0	-149516.1537	2317049.2965	-1496020
F3	DSlbS6	Combination	Bottom	19369.933	1119.048	0	-149736.2964	2594953.2995	-1664084
F3	DSlbS7	Combination	Top	17293.762	0	-1117.815	-97110.5565	2323425.2906	-1502396
F3	DSlbS7	Combination	Bottom	19369.933	0	-1119.048	-97217.7165	2613632.0418	-1682763
F3	DSlbS8	Combination	Top	17293.762	0	1117.815	97110.5565	2310673.3024	-1502396
F3	DSlbS8	Combination	Bottom	19369.933	0	1119.048	97217.7165	2576274.5573	-1682763
F3	DSlbS9	Combination	Top	10641.225	-1117.815	0	149516.1537	1425841.8449	-930832.4301
F3	DSlbS9	Combination	Bottom	11917.779	-1119.048	0	149736.2964	1596825.0476	-1054036
F3	DSlbS10	Combination	Top	10641.225	1117.815	0	-149516.1537	1425841.8449	-918080.4419

Table 3.3 - Story Forces (continued)

Story	Output Case	Case Type	Location	P kip	VX kip	VY kip	T kip-ft	MX kip-ft	MY kip-ft
F3	DSlbS10	Combination	Bottom	11917.779	1119.048	0	-149736.2964	1596825.0476	-1016678
F3	DSlbS11	Combination	Top	10641.225	0	-1117.815	-97110.5565	1432217.8389	-924456.436
F3	DSlbS11	Combination	Bottom	11917.779	0	-1119.048	-97217.7165	1615503.7899	-1035357
F3	DSlbS12	Combination	Top	10641.225	0	1117.815	97110.5565	1419465.8508	-924456.436
F3	DSlbS12	Combination	Bottom	11917.779	0	1119.048	97217.7165	1578146.3053	-1035357
F3	DConS1	Combination	Top	18622.144	0	0	0	2495223.2285	-1617799
F3	DConS1	Combination	Bottom	20856.113	0	0	0	2794443.8333	-1811875
F3	DConS2	Combination	Top	15964.672	0	0	0	2138852.845	-1386931
F3	DConS2	Combination	Bottom	17882.336	0	0	0	2395417.7269	-1553528
F3	DConS3	Combination	Top	17293.762	-1117.815	0	149516.1537	2317049.2965	-1508772
F3	DConS3	Combination	Bottom	19369.933	-1119.048	0	149736.2964	2594953.2995	-1701442
F3	DConS4	Combination	Top	17293.762	1117.815	0	-149516.1537	2317049.2965	-1496020
F3	DConS4	Combination	Bottom	19369.933	1119.048	0	-149736.2964	2594953.2995	-1664084
F3	DConS5	Combination	Top	17293.762	0	-1117.815	-97110.5565	2323425.2906	-1502396
F3	DConS5	Combination	Bottom	19369.933	0	-1119.048	-97217.7165	2613632.0418	-1682763
F3	DConS6	Combination	Top	17293.762	0	1117.815	97110.5565	2310673.3024	-1502396
F3	DConS6	Combination	Bottom	19369.933	0	1119.048	97217.7165	2576274.5573	-1682763
F3	DConS7	Combination	Top	10641.225	-1117.815	0	149516.1537	1425841.8449	-930832.4301
F3	DConS7	Combination	Bottom	11917.779	-1119.048	0	149736.2964	1596825.0476	-1054036
F3	DConS8	Combination	Top	10641.225	1117.815	0	-149516.1537	1425841.8449	-918080.4419
F3	DConS8	Combination	Bottom	11917.779	1119.048	0	-149736.2964	1596825.0476	-1016678
F3	DConS9	Combination	Top	10641.225	0	-1117.815	-97110.5565	1432217.8389	-924456.436
F3	DConS9	Combination	Bottom	11917.779	0	-1119.048	-97217.7165	1615503.7899	-1035357

Table 3.3 - Story Forces (continued)

Story	Output Case	Case Type	Location	P kip	VX kip	VY kip	T kip-ft	MX kip-ft	MY kip-ft
F3	DConS10	Combination	Top	10641.225	0	1117.815	97110.5565	1419465.8508	-924456.436
F3	DConS10	Combination	Bottom	11917.779	0	1119.048	97217.7165	1578146.3053	-1035357
F3	DWalS1	Combination	Top	18622.144	0	0	0	2495223.2285	-1617799
F3	DWalS1	Combination	Bottom	20856.113	0	0	0	2794443.8333	-1811875
F3	DWalS2	Combination	Top	15964.672	0	0	0	2138852.845	-1386931
F3	DWalS2	Combination	Bottom	17882.336	0	0	0	2395417.7269	-1553528
F3	DWalS3	Combination	Top	17293.762	-1117.815	0	149516.1537	2317049.2965	-1508772
F3	DWalS3	Combination	Bottom	19369.933	-1119.048	0	149736.2964	2594953.2995	-1701442
F3	DWalS4	Combination	Top	17293.762	1117.815	0	-149516.1537	2317049.2965	-1496020
F3	DWalS4	Combination	Bottom	19369.933	1119.048	0	-149736.2964	2594953.2995	-1664084
F3	DWalS5	Combination	Top	17293.762	0	-1117.815	-97110.5565	2323425.2906	-1502396
F3	DWalS5	Combination	Bottom	19369.933	0	-1119.048	-97217.7165	2613632.0418	-1682763
F3	DWalS6	Combination	Top	17293.762	0	1117.815	97110.5565	2310673.3024	-1502396
F3	DWalS6	Combination	Bottom	19369.933	0	1119.048	97217.7165	2576274.5573	-1682763
F3	DWalS7	Combination	Top	10641.225	-1117.815	0	149516.1537	1425841.8449	-930832.4301
F3	DWalS7	Combination	Bottom	11917.779	-1119.048	0	149736.2964	1596825.0476	-1054036
F3	DWalS8	Combination	Top	10641.225	1117.815	0	-149516.1537	1425841.8449	-918080.4419
F3	DWalS8	Combination	Bottom	11917.779	1119.048	0	-149736.2964	1596825.0476	-1016678
F3	DWalS9	Combination	Top	10641.225	0	-1117.815	-97110.5565	1432217.8389	-924456.436
F3	DWalS9	Combination	Bottom	11917.779	0	-1119.048	-97217.7165	1615503.7899	-1035357
F3	DWalS10	Combination	Top	10641.225	0	1117.815	97110.5565	1419465.8508	-924456.436
F3	DWalS10	Combination	Bottom	11917.779	0	1119.048	97217.7165	1578146.3053	-1035357
F2	Dead	LinStatic	Top	21745.851	0	0	0	2912306.8799	-1889171

Table 3.3 - Story Forces (continued)

Story	Output Case	Case Type	Location	P kip	VX kip	VY kip	T kip-ft	MX kip-ft	MY kip-ft
F2	Dead	LinStatic	Bottom	23341.543	0	0	0	3126035.8834	-2027797
F2	Live	LinStatic	Top	765.033	0	0	0	102541.2028	-66462.2226 -
F2	Live	LinStatic	Bottom	766.804	0	0	0	102597.5014	66616.0984
F2	Water Pressure	LinStatic	Top	0	0	0	0	0	0
F2	Water Pressure	LinStatic	Bottom	0	0	0	0	0	0
F2	Soil Pressure	LinStatic	Top	0	0	0	0	0	0
F2	Soil Pressure	LinStatic	Bottom	0	0	0	0	0	0
F2	EQ X	LinStatic	Top	0	-1526.766	0	204213.6704	0	-18678.7423
F2	EQ X	LinStatic	Bottom	0	-1527.669	0	204374.7505 -	0	-35478.1364
F2	EQ Y	LinStatic	Top	0	0	-1526.766	132638.215 -	18678.7423	0
F2	EQ Y	LinStatic	Bottom	0	0	-1527.669	132716.6248	35478.1364	0
F2	DSIbS1	Combination	Top	30444.191	0	0	0	4077229.6319	-2644839
F2	DSIbS1	Combination	Bottom	32678.16	0	0	0	4376450.2367	-2838915
F2	DSIbS2	Combination	Top	27319.073	0	0	0	3658834.1804	-2373345
F2	DSIbS2	Combination	Bottom	29236.738	0	0	0	3915399.0623	-2539942
F2	DSIbS3	Combination	Top	30444.191	0	0	0	4077229.6319	-2644839
F2	DSIbS3	Combination	Bottom	32678.16	0	0	0	4376450.2367	-2838915
F2	DSIbS4	Combination	Top	27319.073	0	0	0	3658834.1804	-2373345
F2	DSIbS4	Combination	Bottom	29236.738	0	0	0	3915399.0623	-2539942
F2	DSIbS5	Combination	Top	29034.639	-1526.766	0	204213.6704	3888540.1467	-2541063
F2	DSIbS5	Combination	Bottom	31110.81	-1527.669	0	204374.7505	4166444.1498	-2738230
F2	DSIbS6	Combination	Top	29034.639	1526.766	0	-204213.6704	3888540.1467	-2503706
F2	DSIbS6	Combination	Bottom	31110.81	1527.669	0	-204374.7505	4166444.1498 -	-2667273
F2	DSIbS7	Combination	Top	29034.639	0	1526.766	-132638.215	3907218.889 -	-2522384
F2	DSIbS7	Combination	Bottom	31110.81	0	1527.669	-132716.6248	4201922.2862	-2702752

Table 3.3 - Story Forces (continued)

Story	Output Case	Case Type	Location	P kip	VX kip	VY kip	T kip-ft	MX kip-ft	MY kip-ft
F2	DSlbS8	Combination	Top	29034.639	0	1526.766	132638.215	3869861.4044	-2522384
F2	DSlbS8	Combination	Bottom	31110.81	0	1527.669	132716.6248	4130966.0134	-2702752
F2	DSlbS9	Combination	Top	17396.681	-1526.766	0	204213.6704	2329845.504	-1530015
F2	DSlbS9	Combination	Bottom	18673.234	-1527.669	0	204374.7505	2500828.7067	-1657715
F2	DSlbS10	Combination	Top	17396.681	1526.766	0	-204213.6704	2329845.504	-1492658
F2	DSlbS10	Combination	Bottom	18673.234	1527.669	0	-204374.7505	2500828.7067	-1586759
F2	DSlbS11	Combination	Top	17396.681	0	-1526.766	-132638.215	2348524.2462	-1511337
F2	DSlbS11	Combination	Bottom	18673.234	0	-1527.669	-132716.6248	2536306.8431	-1622237
F2	DSlbS12	Combination	Top	17396.681	0	1526.766	132638.215	2311166.7617	-1511337
F2	DSlbS12	Combination	Bottom	18673.234	0	1527.669	132716.6248	2465350.5703	-1622237
F2	DConS1	Combination	Top	30444.191	0	0	0	4077229.6319	-2644839
F2	DConS1	Combination	Bottom	32678.16	0	0	0	4376450.2367	-2838915
F2	DConS2	Combination	Top	27319.073	0	0	0	3658834.1804	-2373345
F2	DConS2	Combination	Bottom	29236.738	0	0	0	3915399.0623	-2539942
F2	DConS3	Combination	Top	29034.639	-1526.766	0	204213.6704	3888540.1467	-2541063
F2	DConS3	Combination	Bottom	31110.81	-1527.669	0	204374.7505	4166444.1498	-2738230
F2	DConS4	Combination	Top	29034.639	1526.766	0	-204213.6704	3888540.1467	-2503706
F2	DConS4	Combination	Bottom	31110.81	1527.669	0	-204374.7505	4166444.1498	-2667273
F2	DConS5	Combination	Top	29034.639	0	-1526.766	-132638.215	3907218.889	-2522384
F2	DConS5	Combination	Bottom	31110.81	0	-1527.669	-132716.6248	4201922.2862	-2702752
F2	DConS6	Combination	Top	29034.639	0	1526.766	132638.215	3869861.4044	-2522384
F2	DConS6	Combination	Bottom	31110.81	0	1527.669	132716.6248	4130966.0134	-2702752
F2	DConS7	Combination	Top	17396.681	-1526.766	0	204213.6704	2329845.504	-1530015

Table 3.3 - Story Forces (continued)

Story	Output Case	Case Type	Location	P kip	VX kip	VY kip	T kip-ft	MX kip-ft	MY kip-ft
F2	DConS7	Combination	Bottom	18673.234	-1527.669	0	204374.7505	2500828.7067	-1657715
F2	DConS8	Combination	Top	17396.681	1526.766	0	-204213.6704	2329845.504	-1492658
F2	DConS8	Combination	Bottom	18673.234	1527.669	0	-204374.7505	2500828.7067	-1586759
F2	DConS9	Combination	Top	17396.681	0	-1526.766	-132638.215	2348524.2462	-1511337
F2	DConS9	Combination	Bottom	18673.234	0	-1527.669	-132716.6248	2536306.8431	-1622237
F2	DConS10	Combination	Top	17396.681	0	1526.766	132638.215	2311166.7617	-1511337
F2	DConS10	Combination	Bottom	18673.234	0	1527.669	132716.6248	2465350.5703	-1622237
F2	DWalS1	Combination	Top	30444.191	0	0	0	4077229.6319	-2644839
F2	DWalS1	Combination	Bottom	32678.16	0	0	0	4376450.2367	-2838915
F2	DWalS2	Combination	Top	27319.073	0	0	0	3658834.1804	-2373345
F2	DWalS2	Combination	Bottom	29236.738	0	0	0	3915399.0623	-2539942
F2	DWalS3	Combination	Top	29034.639	-1526.766	0	204213.6704	3888540.1467	-2541063
F2	DWalS3	Combination	Bottom	31110.81	-1527.669	0	204374.7505	4166444.1498	-2738230
F2	DWalS4	Combination	Top	29034.639	1526.766	0	-204213.6704	3888540.1467	-2503706
F2	DWalS4	Combination	Bottom	31110.81	1527.669	0	-204374.7505	4166444.1498	-2667273
F2	DWalS5	Combination	Top	29034.639	0	-1526.766	-132638.215	3907218.889	-2522384
F2	DWalS5	Combination	Bottom	31110.81	0	-1527.669	-132716.6248	4201922.2862	-2702752
F2	DWalS6	Combination	Top	29034.639	0	1526.766	132638.215	3869861.4044	-2522384
F2	DWalS6	Combination	Bottom	31110.81	0	1527.669	132716.6248	4130966.0134	-2702752
F2	DWalS7	Combination	Top	17396.681	-1526.766	0	204213.6704	2329845.504	-1530015
F2	DWalS7	Combination	Bottom	18673.234	-1527.669	0	204374.7505	2500828.7067	-1657715
F2	DWalS8	Combination	Top	17396.681	1526.766	0	-204213.6704	2329845.504	-1492658
F2	DWalS8	Combination	Bottom	18673.234	1527.669	0	-204374.7505	2500828.7067	-1586759

Table 3.3 - Story Forces (continued)

Story	Output Case	Case Type	Location	P kip	VX kip	VY kip	T kip-ft	MX kip-ft	MY kip-ft
F2	DWalS9	Combination	Top	17396.681	0	-1526.766	-132638.215	2348524.2462	-1511337
F2	DWalS9	Combination	Bottom	18673.234	0	-1527.669	-132716.6248	2536306.8431	-1622237
F2	DWalS10	Combination	Top	17396.681	0	1526.766	132638.215	2311166.7617	-1511337
F2	DWalS10	Combination	Bottom	18673.234	0	1527.669	132716.6248	2465350.5703	-1622237
F1	Dead	LinStatic	Top	30528.61	0	0	04087835.2785	-2652173	
F1	Dead	LinStatic	Bottom	32259.719	0	0	04318875.6918	-2802157	
F1	Live	LinStatic	Top	766.804	0	0	0102597.5014	-66616.0984	
F1	Live	LinStatic	Bottom	768.647	0	0	0102656.0887	-66776.2295	
F1	Water Pressure	LinStatic	Top	0	0	0	0	0	0
F1	Water Pressure	LinStatic	Bottom	0	0	0	0	0	0
F1	Soil Pressure	LinStatic	Top	0	0	0	0	0	0
F1	Soil Pressure	LinStatic	Bottom	0	0	0	0	0	0
F1	EQ X	LinStatic	Top	0	-1779.959	0	238056.2319	0	-35478.1364
F1	EQ X	LinStatic	Bottom	0	-1780.545	0	238160.8433	0	-56841.0638
F1	EQ Y	LinStatic	Top	0	0	-1779.959	-154624.5012	35478.1364	0
F1	EQ Y	LinStatic	Bottom	0	0	-1780.545	-154675.4464	56841.0638	0
F1	DSlbS1	Combination	Top	42740.055	0	0	0	5722969.39	-3713042
F1	DSlbS1	Combination	Bottom	45163.607	0	0	06046425.9685	-3923020	
F1	DSlbS2	Combination	Top	37861.219	0	0	05069558.3365	-3289193	
F1	DSlbS2	Combination	Bottom	39941.499	0	0	0	5346900.572	-3469431
F1	DSlbS3	Combination	Top	42740.055	0	0	0	5722969.39	-3713042
F1	DSlbS3	Combination	Bottom	45163.607	0	0	06046425.9685	-3923020	
F1	DSlbS4	Combination	Top	37861.219	0	0	05069558.3365	-3289193	
F1	DSlbS4	Combination	Bottom	39941.499	0	0	0	5346900.572	-3469431
F1	DSlbS5	Combination	Top	40453.998	-1779.959	0	238056.2319	5416783.3635	-3549919

Table 3.3 - Story Forces (continued)

Story	Output Case	Case Type	Location	P kip	VX kip	VY kip	T kip-ft	MX kip-ft	MY kip-ft
F1	DSlbS5	Combination	Bottom	42706.282	-1780.545	0	238160.8433	5717194.488	-3766422
F1	DSlbS6	Combination	Top	40453.998	1779.959	0	-238056.2319	5416783.3635	-3478963
F1	DSlbS6	Combination	Bottom	42706.282	1780.545	0	-238160.8433	5717194.488	-3652740
F1	DSlbS7	Combination	Top	40453.998	0	-1779.959	-154624.5012	5452261.4999	-3514441
F1	DSlbS7	Combination	Bottom	42706.282	0	-1780.545	-154675.4464	5774035.5517	-3709581
F1	DSlbS8	Combination	Top	40453.998	0	1779.959	154624.5012	5381305.2271	-3514441
F1	DSlbS8	Combination	Bottom	42706.282	0	1780.545	154675.4464	5660353.4242	-3709581
F1	DSlbS9	Combination	Top	24422.888	-1779.959	0	238056.2319	3270268.2228	-2157217
F1	DSlbS9	Combination	Bottom	25807.775	-1780.545	0	238160.8433	3455100.5534	-2298567
F1	DSlbS10	Combination	Top	24422.888	1779.959	0	-238056.2319	3270268.2228	-2086260
F1	DSlbS10	Combination	Bottom	25807.775	1780.545	0	-238160.8433	3455100.5534	-2184885
F1	DSlbS11	Combination	Top	24422.888	0	-1779.959	-154624.5012	3305746.3593	-2121738
F1	DSlbS11	Combination	Bottom	25807.775	0	-1780.545	-154675.4464	3511941.6172	-2241726
F1	DSlbS12	Combination	Top	24422.888	0	1779.959	154624.5012	3234790.0864	-2121738
F1	DSlbS12	Combination	Bottom	25807.775	0	1780.545	154675.4464	3398259.4897	-2241726
F1	DConS1	Combination	Top	42740.055	0	0	0	5722969.39	-3713042
F1	DConS1	Combination	Bottom	45163.607	0	0	0	6046425.9685	-3923020
F1	DConS2	Combination	Top	37861.219	0	0	0	5069558.3365	-3289193
F1	DConS2	Combination	Bottom	39941.499	0	0	0	5346900.572	-3469431
F1	DConS3	Combination	Top	40453.998	-1779.959	0	238056.2319	5416783.3635	-3549919
F1	DConS3	Combination	Bottom	42706.282	-1780.545	0	238160.8433	5717194.488	-3766422
F1	DConS4	Combination	Top	40453.998	1779.959	0	-238056.2319	5416783.3635	-3478963
F1	DConS4	Combination	Bottom	42706.282	1780.545	0	-238160.8433	5717194.488	-3652740

Table 3.3 - Story Forces (continued)

Story	Output Case	Case Type	Location	P kip	VX kip	VY kip	T kip-ft	MX kip-ft	MY kip-ft
F1	DConS5	Combination	Top	40453.998	0	-1779.959	-154624.5012	5452261.4999	-3514441
F1	DConS5	Combination	Bottom	42706.282	0	-1780.545	-154675.4464	5774035.5517	-3709581
F1	DConS6	Combination	Top	40453.998	0	1779.959	154624.5012	5381305.2271	-3514441
F1	DConS6	Combination	Bottom	42706.282	0	1780.545	154675.4464	5660353.4242	-3709581
F1	DConS7	Combination	Top	24422.888	-1779.959	0	238056.2319	3270268.2228	-2157217
F1	DConS7	Combination	Bottom	25807.775	-1780.545	0	238160.8433	3455100.5534	-2298567
F1	DConS8	Combination	Top	24422.888	1779.959	0	-238056.2319	3270268.2228	-2086260
F1	DConS8	Combination	Bottom	25807.775	1780.545	0	-238160.8433	3455100.5534	-2184885
F1	DConS9	Combination	Top	24422.888	0	-1779.959	-154624.5012	3305746.3593	-2121738
F1	DConS9	Combination	Bottom	25807.775	0	-1780.545	-154675.4464	3511941.6172	-2241726
F1	DConS10	Combination	Top	24422.888	0	1779.959	154624.5012	3234790.0864	-2121738
F1	DConS10	Combination	Bottom	25807.775	0	1780.545	154675.4464	3398259.4897	-2241726
F1	DWalS1	Combination	Top	42740.055	0	0	0	5722969.39	-3713042
F1	DWalS1	Combination	Bottom	45163.607	0	0	0	6046425.9685	-3923020
F1	DWalS2	Combination	Top	37861.219	0	0	0	5069558.3365	-3289193
F1	DWalS2	Combination	Bottom	39941.499	0	0	0	5346900.572	-3469431
F1	DWalS3	Combination	Top	40453.998	-1779.959	0	238056.2319	5416783.3635	-3549919
F1	DWalS3	Combination	Bottom	42706.282	-1780.545	0	238160.8433	5717194.488	-3766422
F1	DWalS4	Combination	Top	40453.998	1779.959	0	-238056.2319	5416783.3635	-3478963
F1	DWalS4	Combination	Bottom	42706.282	1780.545	0	-238160.8433	5717194.488	-3652740
F1	DWalS5	Combination	Top	40453.998	0	-1779.959	-154624.5012	5452261.4999	-3514441
F1	DWalS5	Combination	Bottom	42706.282	0	-1780.545	-154675.4464	5774035.5517	-3709581
F1	DWalS6	Combination	Top	40453.998	0	1779.959	154624.5012	5381305.2271	-3514441

Table 3.3 - Story Forces (continued)

Story	Output Case	Case Type	Location	P kip	VX kip	VY kip	T kip-ft	MX kip-ft	MY kip-ft
F1	DWalS6	Combination	Bottom	42706.282	0	1780.545	154675.4464	5660353.4242	-3709581
F1	DWalS7	Combination	Top	24422.888	-1779.959	0	238056.2319	3270268.2228	-2157217
F1	DWalS7	Combination	Bottom	25807.775	-1780.545	0	238160.8433	3455100.5534	-2298567
F1	DWalS8	Combination	Top	24422.888	1779.959	0	-238056.2319	3270268.2228	-2086260
F1	DWalS8	Combination	Bottom	25807.775	1780.545	0	-238160.8433	3455100.5534	-2184885
F1	DWalS9	Combination	Top	24422.888	0	-1779.959	-154624.5012	3305746.3593	-2121738
F1	DWalS9	Combination	Bottom	25807.775	0	-1780.545	-154675.4464	3511941.6172	-2241726
F1	DWalS10	Combination	Top	24422.888	0	1779.959	154624.5012	3234790.0864	-2121738
F1	DWalS10	Combination	Bottom	25807.775	0	1780.545	154675.4464	3398259.4897	-2241726
Plinth Level	Dead	LinStatic	Top	39124.39	0	0	05238047.1304	-3398526	
Plinth Level	Dead	LinStatic	Bottom	39873.64	0	0	05342161.5446	-3463617	
Plinth Level	Live	LinStatic	Top	1539.378	0	0	0206752.7143	-133733.4908	
Plinth Level	Live	LinStatic	Bottom	1539.378	0	0	0206752.7143	-133733.4908	
Plinth Level	Water Pressure	LinStatic	Top	0	0	0	0	0	0
Plinth Level	Water Pressure	LinStatic	Bottom	0	0	0	0	0	0
Plinth Level	Soil Pressure	LinStatic	Top	0	0	0	0	0	0
Plinth Level	Soil Pressure	LinStatic	Bottom	0	0	0	0	0	0
Plinth Level	EQ X	LinStatic	Top	0	-1869.797	0	250123.043	0	-56841.0638
Plinth Level	EQ X	LinStatic	Bottom	0	-1870.27	0	250234.3809	0	-66191.232
Plinth Level	EQ Y	LinStatic	Top	0	0	-1869.797	-162425.5467	56841.0638	0
Plinth Level	EQ Y	LinStatic	Bottom	0	0	-1870.27	-162466.6563	66191.232	0
Plinth Level	DSlbS1	Combination	Top	54774.146	0	0	07333265.9825	-4757936	
Plinth Level	DSlbS1	Combination	Bottom	55823.096	0	0	07479026.1625	-4849063	
Plinth Level	DSlbS2	Combination	Top	49412.273	0	0	06616460.8994	-4292204	

Table 3.3 - Story Forces (continued)

Story	Output Case	Case Type	Location	P kip	VX kip	VY kip	T kip-ft	MX kip-ft	MY kip-ft
Plinth Level	DSlbS2	Combination	Bottom	50311.373	0	0	0	6741398.1965	-4370314
Plinth Level	DSlbS3	Combination	Top	54774.146	0	0	0	7333265.9825	-4757936
Plinth Level	DSlbS3	Combination	Bottom	55823.096	0	0	0	7479026.1625	-4849063
Plinth Level	DSlbS4	Combination	Top	49412.273	0	0	0	6616460.8994	-4292204
Plinth Level	DSlbS4	Combination	Bottom	50311.373	0	0	0	6741398.1965	-4370314
Plinth Level	DSlbS5	Combination	Top	52401.085	-1869.797	0	250123.043	7016213.9838	-4608658
Plinth Level	DSlbS5	Combination	Bottom	53375.11	-1870.27	0	250234.3809	7151562.7224	-4702626
Plinth Level	DSlbS6	Combination	Top	52401.085	1869.797	0	-250123.043	7016213.9838	-4494976
Plinth Level	DSlbS6	Combination	Bottom	53375.11	1870.27	0	-250234.3809	7151562.7224	-4570244
Plinth Level	DSlbS7	Combination	Top	52401.085	0	1869.797	-162425.5467	7073055.0476	-4551817
Plinth Level	DSlbS7	Combination	Bottom	53375.11	0	1870.27	-162466.6563	7217753.9543	-4636435
Plinth Level	DSlbS8	Combination	Top	52401.085	0	1869.797	162425.5467	6959372.9201	-4551817
Plinth Level	DSlbS8	Combination	Bottom	53375.11	0	1870.27	162466.6563	7085371.4904	-4636435
Plinth Level	DSlbS9	Combination	Top	31299.512	-1869.797	0	250123.043	4190437.7043	-2775662
Plinth Level	DSlbS9	Combination	Bottom	31898.912	-1870.27	0	250234.3809	4273729.2357	-2837085
Plinth Level	DSlbS10	Combination	Top	31299.512	1869.797	0	-250123.043	4190437.7043	-2661979
Plinth Level	DSlbS10	Combination	Bottom	31898.912	1870.27	0	-250234.3809	4273729.2357	-2704702
Plinth Level	DSlbS11	Combination	Top	31299.512	0	1869.797	-162425.5467	4247278.7681	-2718820
Plinth Level	DSlbS11	Combination	Bottom	31898.912	0	1870.27	-162466.6563	4339920.4677	-2770893
Plinth Level	DSlbS12	Combination	Top	31299.512	0	1869.797	162425.5467	4133596.6405	-2718820
Plinth Level	DSlbS12	Combination	Bottom	31898.912	0	1870.27	162466.6563	4207538.0038	-2770893
Plinth Level	DConS1	Combination	Top	54774.146	0	0	0	7333265.9825	-4757936
Plinth Level	DConS1	Combination	Bottom	55823.096	0	0	0	7479026.1625	-4849063

Table 3.3 - Story Forces (continued)

Story	Output Case	Case Type	Location	P kip	VX kip	VY kip	T kip-ft	MX kip-ft	MY kip-ft
Plinth Level	DConS2	Combination	Top	49412.273	0	0	0	6616460.8994	-4292204
Plinth Level	DConS2	Combination	Bottom	50311.373	0	0	0	6741398.1965	-4370314
Plinth Level	DConS3	Combination	Top	52401.085	-1869.797	0	250123.043	7016213.9838	-4608658
Plinth Level	DConS3	Combination	Bottom	53375.11	-1870.27	0	250234.3809	7151562.7224	-4702626
Plinth Level	DConS4	Combination	Top	52401.085	1869.797	0	-250123.043	7016213.9838	-4494976
Plinth Level	DConS4	Combination	Bottom	53375.11	1870.27	0	-250234.3809	7151562.7224	-4570244
Plinth Level	DConS5	Combination	Top	52401.085	0	1869.797	-162425.5467	7073055.0476	-4551817
Plinth Level	DConS5	Combination	Bottom	53375.11	0	1870.27	-162466.6563	7217753.9543	-4636435
Plinth Level	DConS6	Combination	Top	52401.085	0	1869.797	162425.5467	6959372.9201	-4551817
Plinth Level	DConS6	Combination	Bottom	53375.11	0	1870.27	162466.6563	7085371.4904	-4636435
Plinth Level	DConS7	Combination	Top	31299.512	-1869.797	0	250123.043	4190437.7043	-2775662
Plinth Level	DConS7	Combination	Bottom	31898.912	-1870.27	0	250234.3809	4273729.2357	-2837085
Plinth Level	DConS8	Combination	Top	31299.512	1869.797	0	-250123.043	4190437.7043	-2661979
Plinth Level	DConS8	Combination	Bottom	31898.912	1870.27	0	-250234.3809	4273729.2357	-2704702
Plinth Level	DConS9	Combination	Top	31299.512	0	1869.797	-162425.5467	4247278.7681	-2718820
Plinth Level	DConS9	Combination	Bottom	31898.912	0	1870.27	-162466.6563	4339920.4677	-2770893
Plinth Level	DConS10	Combination	Top	31299.512	0	1869.797	162425.5467	4133596.6405	-2718820
Plinth Level	DConS10	Combination	Bottom	31898.912	0	1870.27	162466.6563	4207538.0038	-2770893
Plinth Level	DWalS1	Combination	Top	54774.146	0	0	0	7333265.9825	-4757936
Plinth Level	DWalS1	Combination	Bottom	55823.096	0	0	0	7479026.1625	-4849063
Plinth Level	DWalS2	Combination	Top	49412.273	0	0	0	6616460.8994	-4292204
Plinth Level	DWalS2	Combination	Bottom	50311.373	0	0	0	6741398.1965	-4370314
Plinth Level	DWalS3	Combination	Top	52401.085	-1869.797	0	250123.043	7016213.9838	-4608658

Table 3.3 - Story Forces (continued)

Story	Output Case	Case Type	Location	P kip	VX kip	VY kip	T kip-ft	MX kip-ft	MY kip-ft
Plinth Level	DWalS3	Combination	Bottom	53375.11	-1870.27	0	250234.3809	7151562.7224	-4702626
Plinth Level	DWalS4	Combination	Top	52401.085	1869.797	0	-250123.043	7016213.9838	-4494976
Plinth Level	DWalS4	Combination	Bottom	53375.11	1870.27	0	-250234.3809	7151562.7224	-4570244
Plinth Level	DWalS5	Combination	Top	52401.085	0	-1869.797	-162425.5467	7073055.0476	-4551817
Plinth Level	DWalS5	Combination	Bottom	53375.11	0	-1870.27	-162466.6563	7217753.9543	-4636435
Plinth Level	DWalS6	Combination	Top	52401.085	0	1869.797	162425.5467	6959372.9201	-4551817
Plinth Level	DWalS6	Combination	Bottom	53375.11	0	1870.27	162466.6563	7085371.4904	-4636435
Plinth Level	DWalS7	Combination	Top	31299.512	-1869.797	0	250123.043	4190437.7043	-2775662
Plinth Level	DWalS7	Combination	Bottom	31898.912	-1870.27	0	250234.3809	4273729.2357	-2837085
Plinth Level	DWalS8	Combination	Top	31299.512	1869.797	0	-250123.043	4190437.7043	-2661979
Plinth Level	DWalS8	Combination	Bottom	31898.912	1870.27	0	-250234.3809	4273729.2357	-2704702
Plinth Level	DWalS9	Combination	Top	31299.512	0	-1869.797	-162425.5467	4247278.7681	-2718820
Plinth Level	DWalS9	Combination	Bottom	31898.912	0	-1870.27	-162466.6563	4339920.4677	-2770893
Plinth Level	DWalS10	Combination	Top	31299.512	0	1869.797	162425.5467	4133596.6405	-2718820
Plinth Level	DWalS10	Combination	Bottom	31898.912	0	1870.27	162466.6563	4207538.0038	-2770893
Under Water Level	Dead	LinStatic	Top	40026.833	0	0	0	5362187.941	-3476925
Under Water Level	Dead	LinStatic	Bottom	40306.583	0	0	0	5399149.3897	-3501229
Under Water Level	Live	LinStatic	Top	1538.742	0	0	0	206602.6716	-133678.1959
Under Water Level	Live	LinStatic	Bottom	1538.742	0	0	0	206602.6716	-133678.1959
Under Water Level	Water Pressure	LinStatic	Top	46.631	0	0	0	10971.7745	-4051.0302
Under Water Level	Water Pressure	LinStatic	Bottom	46.631	0	0	0	10971.7745	-4051.0302
Under Water Level	Soil Pressure	LinStatic	Top	-26.154	0	0	0	-6153.6862	2272.0839
Under Water Level	Soil Pressure	LinStatic	Bottom	-26.154	0	0	0	-6153.6862	2272.0839

Table 3.3 - Story Forces (continued)

Story	Output Case	Case Type	Location	P kip	VX kip	VY kip	T kip-ft	MX kip-ft	MY kip-ft
Under Water Level	EQ X	LinStatic	Top	-0.002	-1873.716	0	250707.1984	-0.4606	-66156.115
Under Water Level	EQ X	LinStatic	Bottom	-0.002	-1873.716	0	250707.1984	-0.4606	-69903.5464
Under Water Level	EQ Y	LinStatic	Top	3.405	0	-1873.716	-162765.9823	67016.323	-295.71
Under Water Level	EQ Y	LinStatic	Bottom	3.405	0	-1873.716	-162765.9823	70763.7544	-295.71
Under Water Level	DSIbS1	Combination	Top	56037.566	0	0	0	7507063.1175	-4867695
Under Water Level	DSIbS1	Combination	Bottom	56429.216	0	0	0	7558809.1456	-4901720
Under Water Level	DSIbS2	Combination	Top	50526.95	0	0	0	6772898.745	-4389042
Under Water Level	DSIbS2	Combination	Bottom	50862.65	0	0	0	6817252.4835	-4418206
Under Water Level	DSIbS3	Combination	Top	56037.566	0	0	0	7507063.1175	-4867695
Under Water Level	DSIbS3	Combination	Bottom	56429.216	0	0	0	7558809.1456	-4901720
Under Water Level	DSIbS4	Combination	Top	50526.95	0	0	0	6772898.745	-4389042
Under Water Level	DSIbS4	Combination	Bottom	50862.65	0	0	0	6817252.4835	-4418206
Under Water Level	DSIbS5	Combination	Top	53594.1	-1873.716	0	250707.1984	7182264.6226	-4721616
Under Water Level	DSIbS5	Combination	Bottom	53957.775	-1873.716	0	250707.1984	7230314.5059	-4756958
Under Water Level	DSIbS6	Combination	Top	53594.104	1873.716	0	-250707.1984	7182265.5438	-4589304
Under Water Level	DSIbS6	Combination	Bottom	53957.779	1873.716	0	-250707.1984	7230315.4271	-4617151
Under Water Level	DSIbS7	Combination	Top	53597.507	0	1873.716	-162765.9823	7249281.4062	-4655756
Under Water Level	DSIbS7	Combination	Bottom	53961.182	0	1873.716	-162765.9823	7301078.721	-4687350
Under Water Level	DSIbS8	Combination	Top	53590.696	0	1873.716	162765.9823	7115248.7602	-4655164
Under Water Level	DSIbS8	Combination	Bottom	53954.371	0	1873.716	162765.9823	7159551.2121	-4686759
Under Water Level	DSIbS9	Combination	Top	32021.464	-1873.716	0	250707.1984	4289749.8923	-2847696
Under Water Level	DSIbS9	Combination	Bottom	32245.264	-1873.716	0	250707.1984	4319319.0512	-2870886
Under Water Level	DSIbS10	Combination	Top	32021.468	1873.716	0	-250707.1984	4289750.8134	-2715384

Table 3.3 - Story Forces (continued)

Story	Output Case	Case Type	Location	P kip	VX kip	VY kip	T kip-ft	MX kip-ft	MY kip-ft
Under Water Level	DSlbS10	Combination	Bottom	32245.268	1873.716	0	-250707.1984	4319319.9724	-2731079
Under Water Level	DSlbS11	Combination	Top	32024.872	0	1873.716	-162765.9823	4356766.6758	-2781836
Under Water Level	DSlbS11	Combination	Bottom	32248.672	0	1873.716	-162765.9823	4390083.2662	-2801279
Under Water Level	DSlbS12	Combination	Top	32018.061	0	1873.716	162765.9823	4222734.0298	-2781245
Under Water Level	DSlbS12	Combination	Bottom	32241.861	0	1873.716	162765.9823	4248555.7574	-2800687
Under Water Level	DConS1	Combination	Top	56037.566	0	0	0	7507063.1175	-4867695
Under Water Level	DConS1	Combination	Bottom	56429.216	0	0	0	7558809.1456	-4901720
Under Water Level	DConS2	Combination	Top	50526.95	0	0	0	6772898.745	-4389042
Under Water Level	DConS2	Combination	Bottom	50862.65	0	0	0	6817252.4835	-4418206
Under Water Level	DConS3	Combination	Top	53594.1	-1873.716	0	250707.1984	7182264.6226	-4721616
Under Water Level	DConS3	Combination	Bottom	53957.775	-1873.716	0	250707.1984	7230314.5059	-4756958
Under Water Level	DConS4	Combination	Top	53594.104	1873.716	0	-250707.1984	7182265.5438	-4589304
Under Water Level	DConS4	Combination	Bottom	53957.779	1873.716	0	-250707.1984	7230315.4271	-4617151
Under Water Level	DConS5	Combination	Top	53597.507	0	1873.716	-162765.9823	7249281.4062	-4655756
Under Water Level	DConS5	Combination	Bottom	53961.182	0	1873.716	-162765.9823	7301078.721	-4687350
Under Water Level	DConS6	Combination	Top	53590.696	0	1873.716	162765.9823	7115248.7602	-4655164
Under Water Level	DConS6	Combination	Bottom	53954.371	0	1873.716	162765.9823	7159551.2121	-4686759
Under Water Level	DConS7	Combination	Top	32021.464	-1873.716	0	250707.1984	4289749.8923	-2847696
Under Water Level	DConS7	Combination	Bottom	32245.264	-1873.716	0	250707.1984	4319319.0512	-2870886
Under Water Level	DConS8	Combination	Top	32021.468	1873.716	0	-250707.1984	4289750.8134	-2715384
Under Water Level	DConS8	Combination	Bottom	32245.268	1873.716	0	-250707.1984	4319319.9724	-2731079
Under Water Level	DConS9	Combination	Top	32024.872	0	1873.716	-162765.9823	4356766.6758	-2781836
Under Water Level	DConS9	Combination	Bottom	32248.672	0	1873.716	-162765.9823	4390083.2662	-2801279

Table 3.3 - Story Forces (continued)

Story	Output Case	Case Type	Location	P kip	VX kip	VY kip	T kip-ft	MX kip-ft	MY kip-ft
Under Water Level	DConS10	Combination	Top	32018.061	0	1873.716	162765.9823	4222734.0298	-2781245
Under Water Level	DConS10	Combination	Bottom	32241.861	0	1873.716	162765.9823	4248555.7574	-2800687
Under Water Level	DWalS1	Combination	Top	56037.566	0	0	0	7507063.1175	-4867695
Under Water Level	DWalS1	Combination	Bottom	56429.216	0	0	0	7558809.1456	-4901720
Under Water Level	DWalS2	Combination	Top	50526.95	0	0	0	6772898.745	-4389042
Under Water Level	DWalS2	Combination	Bottom	50862.65	0	0	0	6817252.4835	-4418206
Under Water Level	DWalS3	Combination	Top	53594.1	-1873.716	0	250707.1984	7182264.6226	-4721616
Under Water Level	DWalS3	Combination	Bottom	53957.775	-1873.716	0	250707.1984	7230314.5059	-4756958
Under Water Level	DWalS4	Combination	Top	53594.104	1873.716	0	-250707.1984	7182265.5438	-4589304
Under Water Level	DWalS4	Combination	Bottom	53957.779	1873.716	0	-250707.1984	7230315.4271	-4617151
Under Water Level	DWalS5	Combination	Top	53597.507	0	1873.716	-162765.9823	7249281.4062	-4655756
Under Water Level	DWalS5	Combination	Bottom	53961.182	0	1873.716	-162765.9823	7301078.721	-4687350
Under Water Level	DWalS6	Combination	Top	53590.696	0	1873.716	162765.9823	7115248.7602	-4655164
Under Water Level	DWalS6	Combination	Bottom	53954.371	0	1873.716	162765.9823	7159551.2121	-4686759
Under Water Level	DWalS7	Combination	Top	32021.464	-1873.716	0	250707.1984	4289749.8923	-2847696
Under Water Level	DWalS7	Combination	Bottom	32245.264	-1873.716	0	250707.1984	4319319.0512	-2870886
Under Water Level	DWalS8	Combination	Top	32021.468	1873.716	0	-250707.1984	4289750.8134	-2715384
Under Water Level	DWalS8	Combination	Bottom	32245.268	1873.716	0	-250707.1984	4319319.9724	-2731079
Under Water Level	DWalS9	Combination	Top	32024.872	0	1873.716	-162765.9823	4356766.6758	-2781836
Under Water Level	DWalS9	Combination	Bottom	32248.672	0	1873.716	-162765.9823	4390083.2662	-2801279
Under Water Level	DWalS10	Combination	Top	32018.061	0	1873.716	162765.9823	4222734.0298	-2781245
Under Water Level	DWalS10	Combination	Bottom	32241.861	0	1873.716	162765.9823	4248555.7574	-2800687

3.3 Modal Results

Table 3.4 - Modal Periods And Frequencies

Case	Mode	Period sec	Frequency cyc/sec	CircFreq rad/sec	Eigenvalue rad2/sec2
Modal	1	0.473	2.116	13.2937	176.7214
Modal	2	0.465	2.151	13.5141	182.6299
Modal	3	0.441	2.268	14.2523	203.1286
Modal	4	0.323	3.097	19.4578	378.6077
Modal	5	0.259	3.862	24.2667	588.8725
Modal	6	0.171	5.857	36.8006	1354.2833
Modal	7	0.146	6.873	43.1834	1864.802
Modal	8	0.145	6.876	43.2033	1866.5234
Modal	9	0.142	7.065	44.3926	1970.7051
Modal	10	0.138	7.233	45.448	2065.5202
Modal	11	0.132	7.554	47.4608	2252.532
Modal	12	0.126	7.954	49.9771	2497.7147